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1. Introduction

The aim of this project is to model the katabatic wind in a two-dimensional valley. A katabatic wind, a type of density-driven flow, is a common and sometimes enormous phenomenon. In Zadar, Croatia in 1998, a record speed of 248 km/h was observed; fresh water froze inside the pipes, trees were ripped out from the soil roofs were destroyed [1]. Prandtl described katabatic wind as ‘a striking phenomenon in all long valleys among the mountains’ [4].

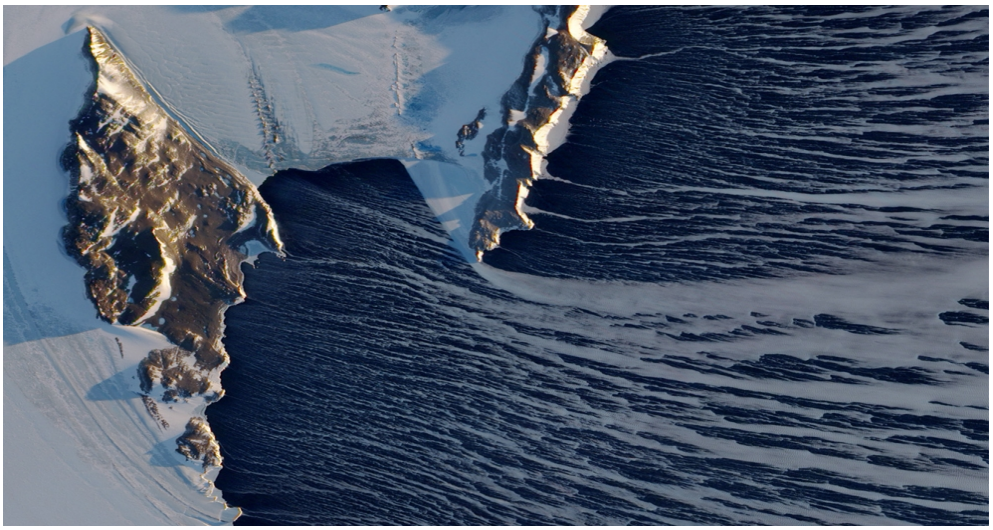


Fig. 1. Katabatic winds at the Antarctic [2]

The physical origin of a katabatic wind is understood by considering the thermodynamics of a mountain slope, as shown in Fig.2. The mountain surface gets heated up by incident radiation during the day and gets cooler as heat is radiated away at night. The diurnal temperature variation of the ground is greater than the ambient air because of larger thermal diffusivity of earth than the air. This means that the ground gets colder than the adjacent air at night, creating a layer of cooler, denser air near the surface. Gravity drags this thermal boundary layer downhill, at faster and faster speeds initially as the ground extracts thermal energy along the way. The flow eventually slows down when the temperature difference with the ground diminishes and kinetic energy is dissipated, or when separate streams of katabatic winds collide to consume their momentum.

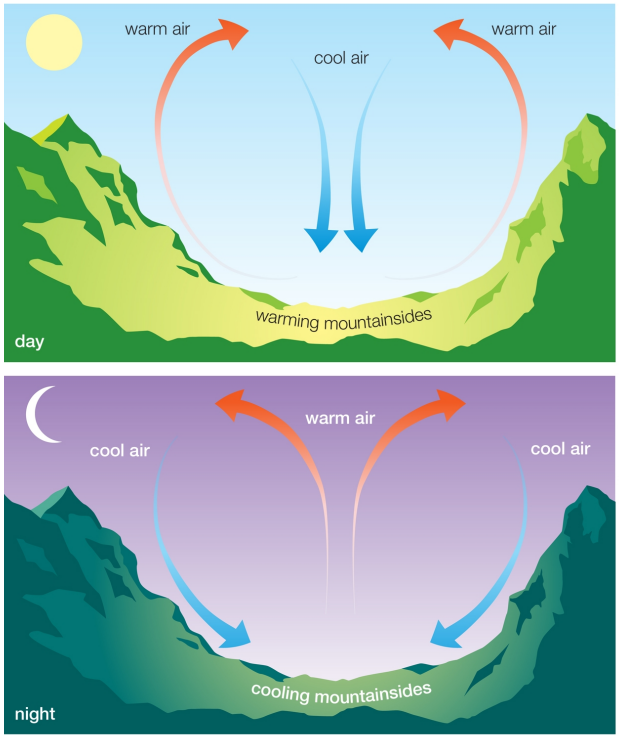


Fig. 2. Diurnal Temperature Variation Driving Flow [5]



Fig. 3. Air Flowing out of Valley [6]

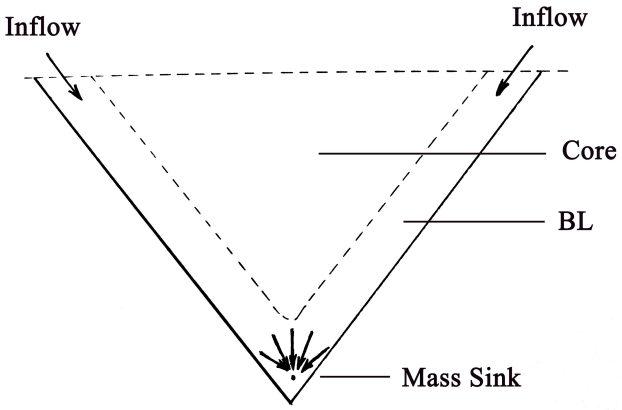


Fig. 4. Cross Section of an Open Valley

It is important to distinguish between an 'open' valley, where flow is allowed to exit the valley, and a 'blocked' valley, where flow is confined within it. Let us consider a long valley with constant V-shaped cross section along its length, so that the katabatic flow stays in the cross-sectional plane and is invariant along the length of the valley. In his model 'Mountain and Valley Winds in Stratified Air' [4], Prandtl assumed the valley to be open

ended, as illustrated by Fig. 3. For a constant slope angle, the wall jet keeps going downslope and simply turns 90º at the bottom of the valley to exit the plane like a river. Fig. 4 is an elevation of the valley cross section showing an inflow at the top which supplies mass flux into the BL, and a sink at the foot of the valley. The core flow is in fact quiescent and the motion is confined to the BL only. Prandtl made the further assumption of an infinite plate, so that the BL thickness, velocity, and temperature difference are all invariant along the slope [4]. These simplifying assumptions result in a model that can be treated analytically.

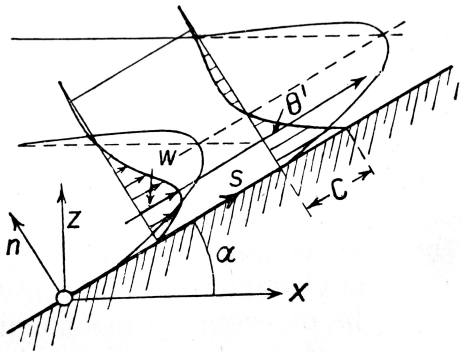


Fig. 5. Prandtl's Model of an Infinite Heated Plate [4]

The focus of this project is on a more complex and challenging case where the valley is blocked on the ends. In this case, the flow cannot exit the cross-sectional plane and have to recirculate back up after reaching the bottom of the valley (Fig. 6). Recirculating flows are very common and more difficult to model.

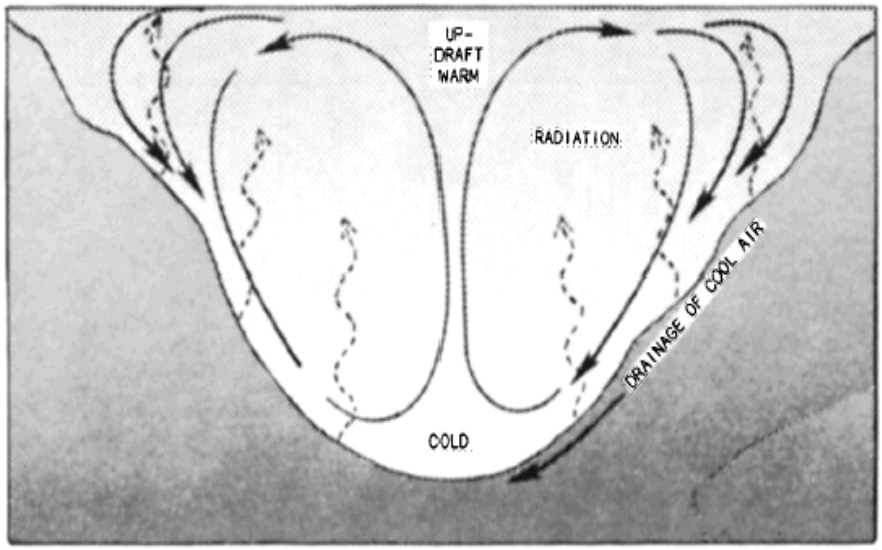


Fig. 6. Recirculating Flow Driven by Density [7]

This project concentrates on deriving the velocity and temperature distributions in the valley. Apart from modelling a more complex flow pattern than the infinite plate model, we use different boundary conditions (B.C's). The assumption of an infinite plate means that the wall is always colder than the adjacent core flow by a fixed temperature, which is unrealistic because the ground cannot cool indefinitely as thermal diffusion would eventually eliminate the temperature difference between the air and the ground. Therefore, in this project the ground is modelled to be at a fixed temperature. Moreover, Prandtl only modelled the flow inside the BL while this project will try to match the core flow to the BL flow using B.C's.

The recirculating flow resembles a model for Aluminum Ingot by Davidson [8], which is developed for an entirely separate purpose. Nevertheless, the derivation of core temperature stratification is adapted in this report. However, the method of solution, the flow pattern, and the conclusions are quite different and based on original work.

Whether the flow is laminar or turbulent also has a profound influence on the method of modelling. While understanding that turbulent flows are ubiquitous and important in the nature, this project has investigated the case for laminar flows. Due to the constraints of time and the complexity of the problem, the only reasonable way to model turbulence is to assume the flow to be laminar and later adjust the kinematic viscosity to an eddy viscosity. Similarly, the laminar thermal diffusivity is replaced by a turbulent diffusivity, assuming that the Prandtl number remains unchanged.

2. Derivation of Governing Equations

2.1. Structure of Analysis

We start analysing the flow by dividing it into a free stream core flow and a BL, as shown in Fig. 7 below. Velocity and temperature gradients are negligible in the core and are significant inside the BL. The Prandtl number for air is close to 1 because the thermal diffusivity and kinematic viscosity of air have very similar magnitudes. Therefore, for simplicity we have taken Pr = 1 in the initial analysis, so that the viscous boundary layer (VBL) and thermal boundary layer (TBL) have a common thickness denoted by δ. Pr is changed to a more accurate value later in the solution process.

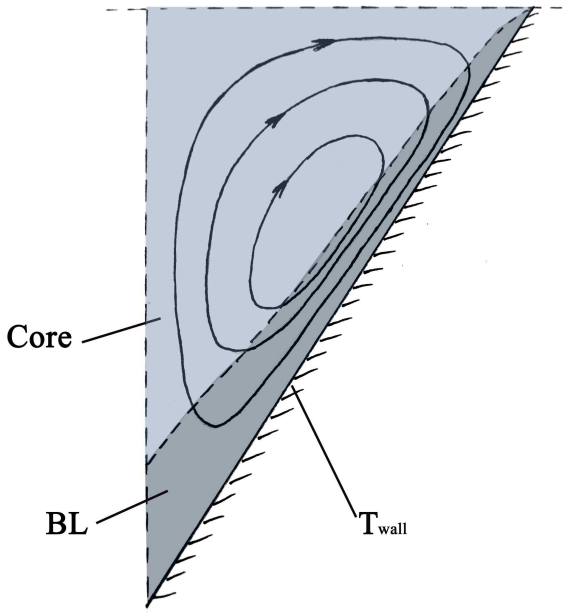


Fig. 7. Half Cross-section: Flow Divided into a Core and a BL

Dividing the flow into a core and a BL enables us to analyse them separately. In the analysis that follows, we will derive the governing equations for each region, combine them, and try to solve them using B.C's that link the BL to the core. Finally, we will examine the solutions to plot the flow and temperature fields throughout the core and the BL.

2.2. Core Flow Stratification

Consider a parcel of air initially at the same temperature with the surrounding air under steady conditions. When the air element loses heat due to radiation, it shrinks in volume under the local pressure. This fractional change in volume due to a change in temperature is known as β, the volumetric thermal expansivity [3]. The isobaric compression subjects the parcel of air to a net buoyancy force of – gβT, which is balanced by the hydrostatic pressure. Since viscosity is negligible in the core, the change in the velocity of a fluid parcel is the result of an imbalance between the buoyancy force and the pressure force. We can therefore invoke the Navier Stokes equation with a buoyancy term:

$$\underset{\text{I}}{\frac{D\mathbf{u}}{Dt}} = \underset{\text{II}}{-\nabla\left(\frac{P}{\rho}\right)} \underset{\text{III}}{- g\beta T\mathbf{k}} \underset{\text{IV}}{+ \nu\nabla^2\mathbf{u}} \tag{1}$$

I: kinetic forces

II: pressure force

III: net buoyancy force

IV: viscous forces

Now consider the transport of heat. Since the air is warmer at a greater altitude and the ground is colder, there is downward heat diffusion balanced by upward advection:

$$\frac{DT}{Dt} = \alpha\nabla^2 T \tag{2}$$

In his model, Prandtl simply assumed that the core flow was thermally stratified, that is, the core temperature distribution is a function of the vertical distance only (Fig. 8). We will verify that this is a physical and mathematical certainty and thereby derive our key equation for the core. Throughout this report we will use an x-z coordinate with z pointing downwards, as labelled in Fig. 8.

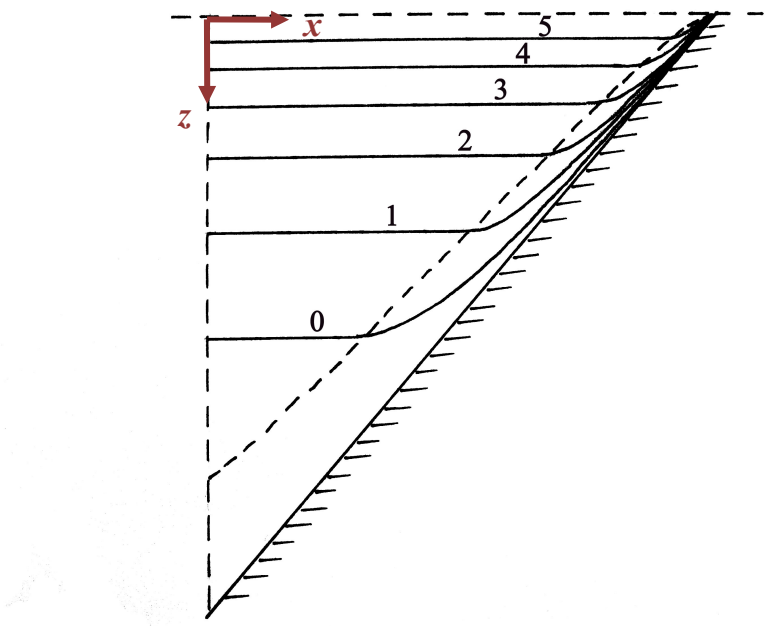


Fig. 8. Characteristic Isotherms due to Thermal Stratification

We start by integrating Eq. (1) around a closed streamline:

$$\oint_C \frac{D\mathbf{u}}{Dt}\cdot d\mathbf{l} = -\oint_C \nabla\left(\frac{P}{\rho}\right)\cdot d\mathbf{l} - g\beta\oint_C T dz + \nu\oint_C \nabla^2\mathbf{u}\cdot d\mathbf{l}$$

Under steady state, $\oint_C \frac{D\mathbf{u}}{Dt}\cdot d\mathbf{l} = \oint_C (\mathbf{u}\cdot\nabla)\mathbf{u}\cdot d\mathbf{l}$

By making use of the vector relationship $(\mathbf{u}\cdot\nabla)\mathbf{u} = \nabla(\frac{1}{2}\mathbf{u}^2) - \mathbf{u}\times(\nabla\times\mathbf{u})$, the RHS can be broken down into two parts:

$$\oint_C \nabla(\frac{1}{2}\mathbf{u}^2)\cdot d\mathbf{l} = \iint_S \nabla\times\nabla(\frac{1}{2}\mathbf{u}^2)\cdot d\mathbf{l} = 0$$

$$\oint_C \mathbf{u}\times(\nabla\times\mathbf{u})\cdot d\mathbf{l} = \oint_C \mathbf{u}\times\boldsymbol{\omega}\cdot d\mathbf{l} = 0$$

where Stokes' Theorem has been applied to derive the first equation. The second equation holds because the cross product, $\mathbf{u}\times\boldsymbol{\omega}$, is normal to $\mathbf{u}$ while $d\mathbf{l}$ is in the same direction as $\mathbf{u}$. We have therefore proven that $\oint_C \frac{D\mathbf{u}}{Dt}\cdot d\mathbf{l} = 0$.

Applying Stokes' Theorem to the pressure term removes it from the equation and we are left with:

$$g\beta\oint_C T dz = \upsilon\oint_C \nabla^2\mathbf{u}\cdot d\mathbf{l} \tag{3}$$

Eq. (3) states that, for every closed streamline, the buoyancy term, which acts as a source for kinetic energy, is balanced by viscous dissipation. Davidson [8] gave the following four solutions to this equation:

A. $\mathbf{u}$ scales as $\frac{1}{\nu}$

B. All streamlines pass through a singular region, where the velocity gradients scale as $\nu^{\frac{1}{2}}$

C.i. There exists a stratified region T = T (z), and

C.ii. Streamlines in this region close within the stratified core and never enter the BL.

D. as in C, but the core is isothermal ($T = T_{wall}$)

Consider the relevance of these solutions to our problem: A is unphysical as ν becomes zero in the core; D states that cross-stream diffusion of heat homogenises temperature, and can be taken as a special case of C. Therefore, our solution must lie between B and C.

Solution B demands that all core streamlines must pass through the BL, where kinetic energy is dissipated. On the other hand, C states that if the core is satisfied according to $T = T(z)$, derivation (see Appendix I) shows that $g\beta\oint_C T dz = 0$, which means that no energy is created or dissipated as we go round the closed streamline.

We now set out to prove that C.i and B must both bold true at the same time. We start by proving that holding C.i true leads to B:

Denoting core properties by the footnote c, substitution of T = T (z) into Eq. (2) gives

$$u_{zc}T_c'(z) = \alpha T_c''(z) \tag{4}$$

where u_{zc} is the core velocity in z-direction. This gives $u_{zc} = \frac{\alpha T_c''(z)}{T_c'(z)}$, which is a function of z only if T = T (z).

Next we invoke continuity to examine the variation of x-direction velocity:

$$\frac{\partial u_{xc}}{\partial x} + \frac{\partial u_{zc}}{\partial z} = 0,\ \frac{\partial u_{zc}}{\partial z} = F(z)$$

$\Rightarrow \frac{\partial u_{xc}}{\partial x} = -F(z)$ and u_{xc} therefore takes the form of $u_{xc} = -xF(z) + \text{Constant}$. Hence u_{xc} is a linear function of x for a fixed z. Furthermore, since symmetry demands that $u_{xc} = 0$ at x = 0, u_{xc} must be non-zero at the edge of the BL, which is to say that all core streamlines will eventually enter the BL, proving the statement in B.

Having proven that C.i leads to B, the next step is to prove that holding B true leads to C.i. This is most easily seen by taking an order-of-magnitude approach.

As heat is conducted to the wall, core temperature drops with depth, asymptoting to T_{wall}, and eventually forms an isothermal region at the foot of the valley. We introduce an important length scale across which temperature decays significantly, L_T. It is the characteristic length of the stratified region (if it exists), and forms the vertical length scale to be used in our order-of-magnitude analysis.

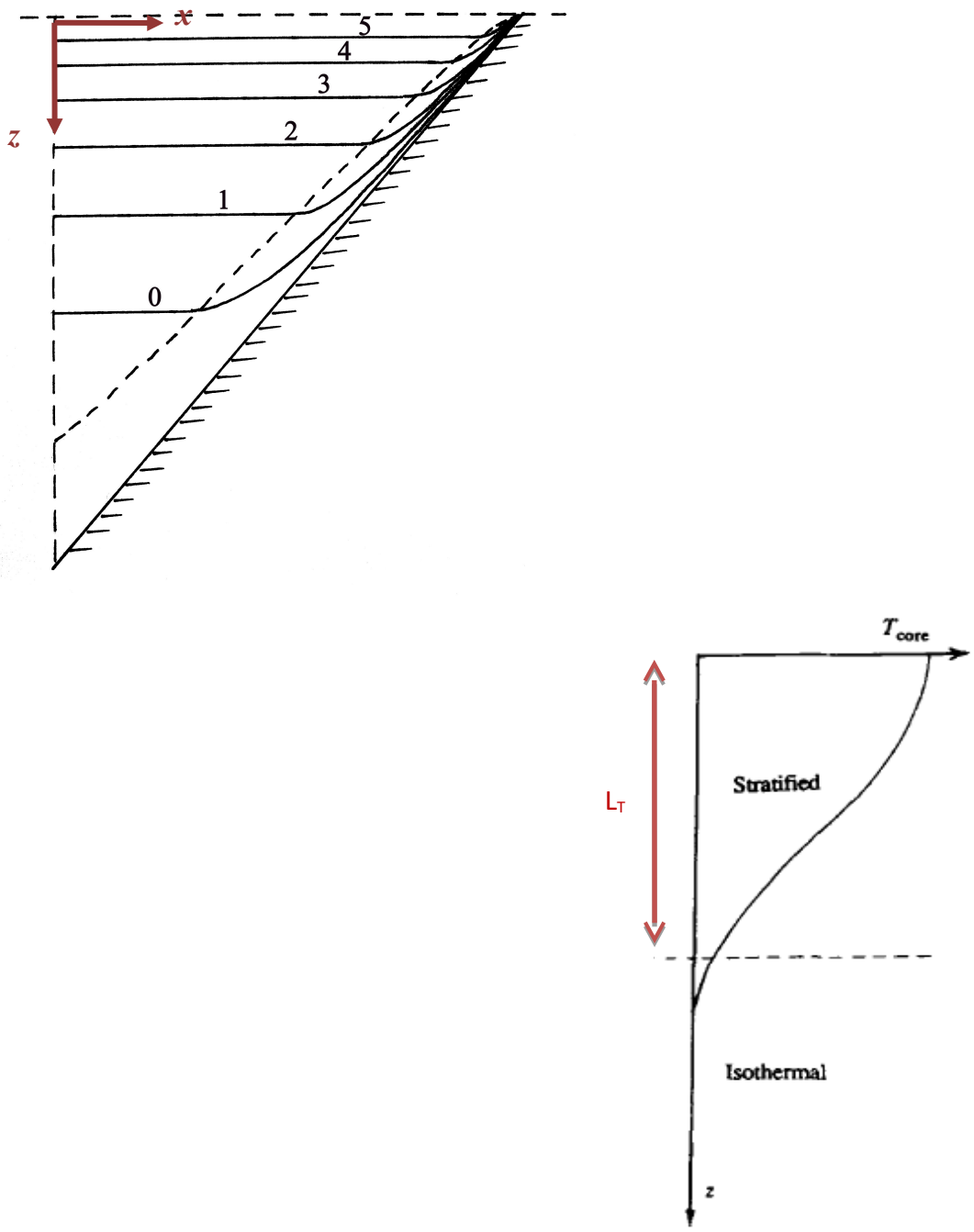


Fig. 9. Stratified Core of Depth L_T above the Isothermal Region

Inside the TBL, competition is initially between the kinetic and the buoyancy forces. Viscous force tries to stop the motion but cannot succeed. The kinetic and the buoyancy terms in Eq. (1) therefore have the same order of magnitude:

$$(\mathbf{u}\cdot\nabla)\mathbf{u} \approx g\beta T\mathbf{k}$$

Taking the core temperature (T_c) as the scale for T, the above is written in scalar form:

$$u_{zb}\frac{\partial u_{zb}}{\partial z} \approx g\beta T_c \Rightarrow \frac{u_{zb}^{\ 2}}{L_T} \approx g\beta T_c \tag{5}$$

As u_{zb} goes downslope, u_{zc} must go upwards, in such a way that continuity is satisfied at any depth of the valley, as illustrated in Fig. 10.

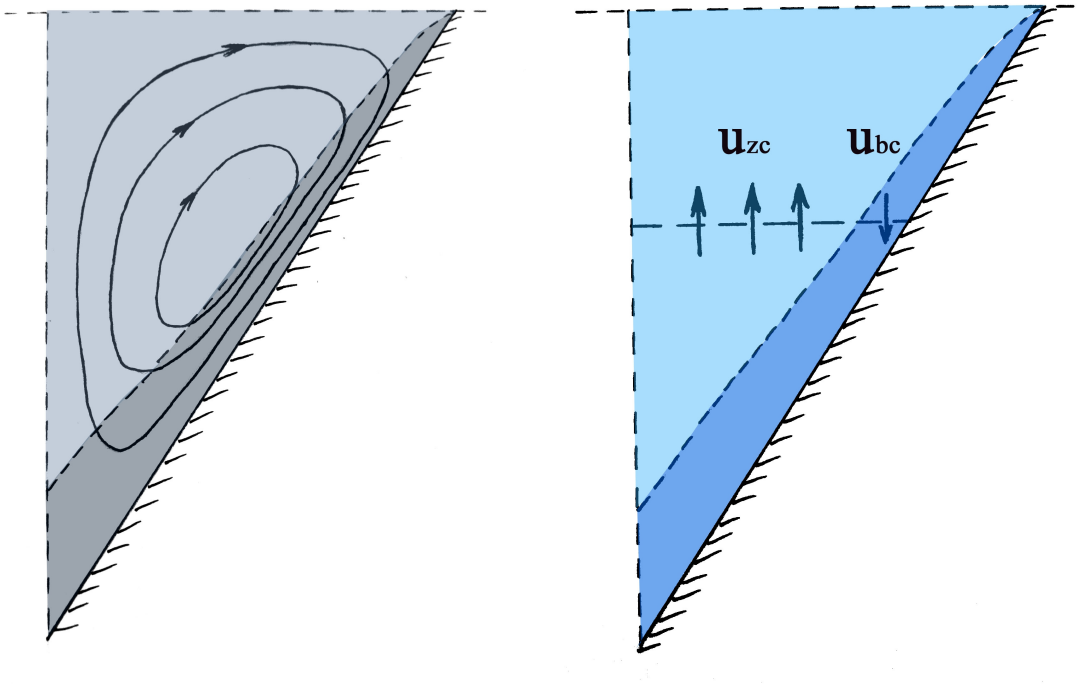


Fig. 10. Continuity at a Horizontal Section

This enables us to relate velocity in the core to that in the BL:

$$u_{zc}(W-\delta) \approx u_{zb}\delta \Rightarrow u_{zc} \approx u_{zb}\frac{\delta}{W} \tag{6}$$

where W is the maximum half-width of the valley. It is assumed that $\frac{\delta}{W} \ll 1$ so that $(W-\delta) \approx W$.

Assuming that $L_T \gg W$ and that $\delta \ll L_T$, an order-of-magnitude analysis given out in Appendix II proves that T_c is a function of z only. The above assumptions will be re-examined and verified in Section 4.3.

In conclusion, we have proven that *all streamlines must pass through the BL, and that the core is thermally stratified*, which gives Eq. (4), the key equation for the core:

$$u_{zc} = \frac{\alpha T_c''(z)}{T_c'(z)} \tag{4}$$

2.3. Boundary Layer Equations

The BL is a region where there are substantial gradients in velocity and temperature; heat and momentum therefore diffuse significantly.

2.3.1. Momentum Integral Equation

Consider the momentum balance inside the BL. Kinetic force, buoyancy, and viscous shear are all important. Stream-wise pressure gradient is also significant, although we can make the reasonable assumption that cross-stream pressure gradient is negligible. We define a slope coordinate system as shown in Fig. 11 with its origin at the tip of the slope. We can use this to examine a fluid element on the slope that spans Δs along the slope and δ in the direction normal to the slope. The mountain surface is modelled to be at a fixed angle ϕ to the horizontal.

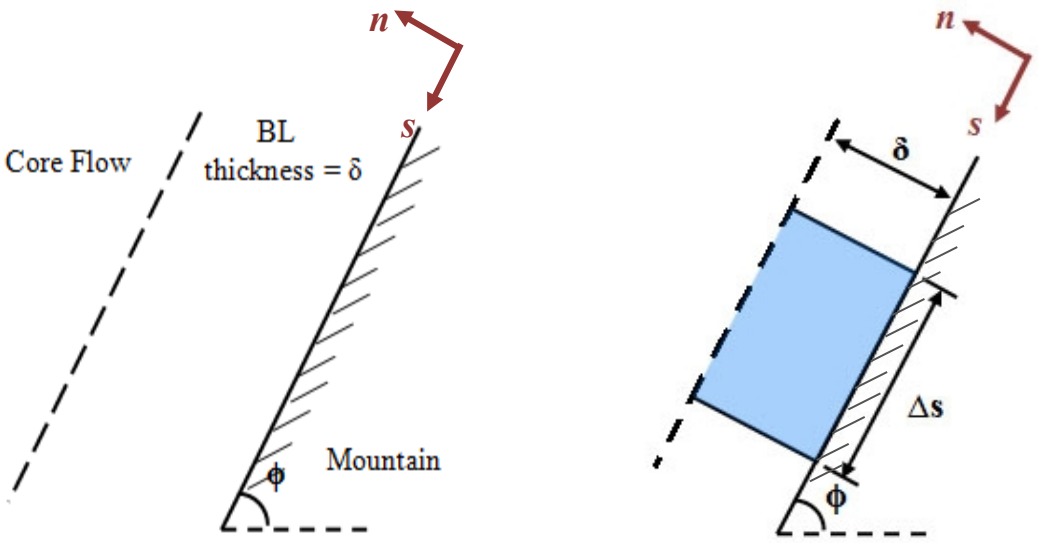


Fig. 11. Fluid Element on the Slope

Integration of Eq. (1) over the closed volume of the fluid element gives

$$\iiint_V \nabla\cdot(u_s\mathbf{u})dV = -\frac{1}{\rho}\iiint_V \frac{\partial p}{\partial s}dV - g\beta\sin\phi\iiint_V T dV + \frac{1}{\rho}\iiint_V \frac{\partial \tau}{\partial n}dV$$

In the above we have assumed that the flow is laminar and modelled the wall shear stress to be $\frac{\tau}{\rho} = \nu\frac{\partial u_s}{\partial n}$, where ν is the kinematic viscosity.

Assuming that u_s adopts a profile that goes to zero at the wall and the BL boundary, we can simplify the volume integrals to derive our key momentum integral equation:

$$\underset{\text{I}}{\frac{d}{ds}\int_0^\delta u_s^2 dn} = \underset{\text{II}}{g\beta\sin\phi}\underset{\text{III}}{\int_0^\delta (T_c - T)dn} - \underset{\text{IV}}{\frac{\tau_w}{\rho}} \qquad (9)$$

I: rate of change of momentum per unit mass

II: acceleration due to pressure force

III: acceleration due to buoyancy force

IV: acceleration due to wall shear

2.3.2. Heat Transport Equation

The transport equation for heat is obtained similarly by integrating Eq. (2) across the fluid volume:

$$\iiint_V \nabla \cdot (\mathbf{u}T)dV = \iiint_V \nabla(\alpha \nabla T)dV$$

Using Stokes' Theorem, the volume integrals are transformed into surface integrals and then expanded. Note that the footnotes s and n denote slope-wise and slope-normal quantities respectively:

$$\iint_S \mathbf{u}T \cdot d\mathbf{A} = \iint_S \alpha \nabla T \cdot d\mathbf{A} \Rightarrow$$

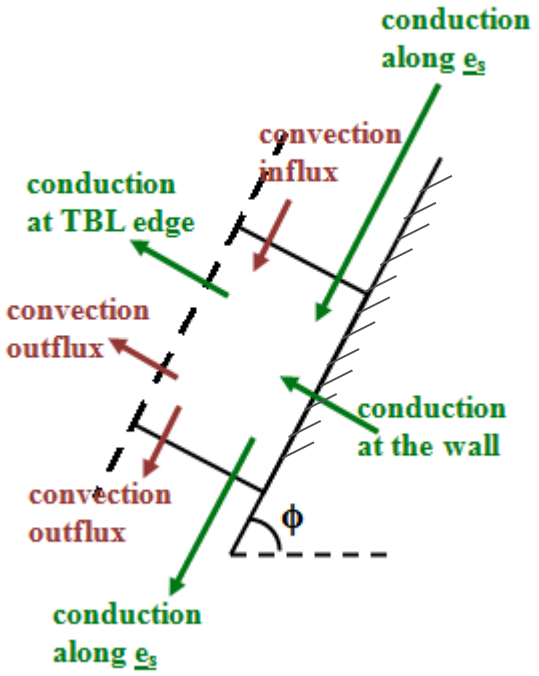


Fig. 12. Heat Balance of a Fluid Element

$$\underset{\text{I}}{\int_0^\delta \frac{\partial(u_s T)}{\partial s}\Delta s dn} + \underset{\text{II}}{\left(u_n T_c\right)_{n=\delta}\Delta s} = \underset{\text{III}}{-\alpha \int_0^\delta \frac{\partial^2 T}{\partial s^2}\Delta s dn} \underset{\text{IV}}{-\alpha \Delta s \left(\frac{\partial T}{\partial n}\right)_{n=0}} + \underset{\text{V}}{\alpha \Delta s \left(\frac{\partial T}{\partial n}\right)_{n=\delta}}$$

I: convection along slope

II: heat advected across BL boundary

III: conduction along slope

IV: conduction at the wall

V: conduction on BL boundary

Further simplification gives our key transport equation for heat:

$$\frac{d}{ds}\int_0^\delta u_s T - T_c \int_0^\delta \frac{\partial u_s}{\partial s} dn = \alpha \left[-\left(\frac{\partial T}{\partial n}\right)_{n=0} + \frac{\partial T_c}{\partial n} \right] \tag{10}$$

2.4. Combining Core and Boundary Layer Equations

We now need to combine the core equation with the BL equations. This is done by enforcing global continuity (recall Fig. 10) and introducing $\dot{q}$, the volumetric flux through the BL:

$$\dot{q} = \int_0^{\delta} u_s dn = -w u_{zc} = -\alpha w \frac{T_c''(z)}{T_c'(z)} \tag{11}$$

where w is the local half-width of the valley and is a linear function of depth (z) for a constant slope angle (ϕ): $w = W\left(1 - \frac{z}{W \tan\phi}\right)$

$\dot{q}$ is our key quantity that connects the BL and core quantities.

2.5. Simplifications to the Combined Equations

2.5.1. Dimensionless Coefficients

To model the wall jet we invoke the assumption of self-similarity, which means that the velocity and temperature profiles keep the same shape as the flow evolves, although their magnitudes vary. This assumption is based on the fact that flows tend to reach equilibrium quickly so that the balance of momentum and temperature terms is not disturbed. This enables us to introduce a set of dimensionless coefficients ($\chi_i's$):

$$\left(\frac{\partial u_s}{\partial n}\right)_{n=0} = \chi_0 \frac{\dot{q}}{\delta^2} \tag{12}$$

$$\left(\frac{\partial T}{\partial n}\right)_{n=0} = \chi_1 \frac{T_c}{\delta} \tag{13}$$

$$\int_0^{\delta} u_s^2 dn = \chi_2 \frac{\dot{q}^2}{\delta} \tag{14}$$

$$\int_0^{\delta}(T_c - T)dn = \chi_3 T_c \delta \tag{15}$$

$$\int_0^{\delta} u_s(T_c - T)dn = \chi_4 T_c \dot{q} \tag{16}$$

These dimensionless coefficients are very useful because they contain all the information we need from the velocity and temperature profiles. Furthermore, self-similarity means that these χ_i's are in fact independent of the distance along the slope, because the velocity and temperature profiles keep the same shape.

The χ_i's also bring great simplifications to solving the equations as they convert the key integral and differential terms in Eq. (9) and (10) to plain forms. Recall the assumption that the thermally stratified region is thin relative to the valley depth ($L_T \ll W$), we confine our analysis to the top region of the valley ($z < L_T$) so that $w \approx W$. Equation (11) then becomes

$\dot{q} \approx -\alpha W \dfrac{T_c''(z)}{T_c'(z)}$. Substitution of $\dot{q}$ and the χ_i's into Eq. (9) and (10) gives:

$$\chi_2 \alpha^2 \frac{d}{ds}\left(\frac{W}{\delta}\left[\frac{T_c''(z)}{T_c'(z)}\right]^2\right) = g\beta \sin\phi \chi_3 T_c \delta - \chi_0 \alpha \frac{\nu W}{\delta^2}\frac{T_c''(z)}{T_c'(z)} \tag{17}$$

$$-\chi_4 \alpha \frac{d}{ds}\left(\frac{T_c''(z)T_c}{T_c'(z)}W\right) + \alpha W \frac{T_c''(z)}{T_c'(z)}\frac{dT_c}{ds} = \alpha \chi_1 \frac{T_c}{\delta} + \alpha \cos\phi \frac{dT_c}{dz} \tag{18}$$

As long as we are able to estimate the values of the χ_i's, Eq. (17) and (18) contain only two unknowns: the core temperature T_c and the BL thickness δ.

To evaluate the χ_i's we introduce two functions that define the BL velocity and temperature profiles:

$$u_s = \frac{\dot{q}}{\delta} h(\eta) \qquad\qquad \frac{T_c - T}{T_c - T_{wall}} = \frac{T_c - T}{T_c} = f(\eta)$$

where $\frac{n}{\delta}=\eta$ and T_{wall} is the datum temperature of the mountain surface and is set to zero for simplification. $h(\eta)$ and $f(\eta)$ are subject to the following B.C's:

$$\int_0^1 u_s d\eta = \frac{\dot{q}}{\delta} \Rightarrow \int_0^1 h(\eta)d\eta = 1 \tag{19}$$

$$\left(\frac{T_c - T}{T_c}\right)_{\eta=0} = 1 \Rightarrow \left[f(\eta)\right]_{\eta=0} = 1 \tag{20}$$

Making use of Eqs. (12) ~ (16), the dimensionless coefficients can be expressed in terms of these profile functions:

$$\chi_0 = -\left[h'(\eta)\right]_{\eta=0} \qquad \chi_1 = -\left[f'(\eta)\right]_{\eta=0}$$

$$\chi_2 = \int_0^1 h^2(\eta)d\eta \qquad \chi_3 = \int_0^1 f(\eta)d\eta$$

$$\chi_4 = \int_0^1 h(\eta)f(\eta)d\eta$$

From the above it is clear that to solve the key momentum and heat equations, we need not know the specific velocity or temperature profiles, only the dimensionless coefficients. By varying the $\chi_i{}'s$ we can input different shapes which are preserved along the slope. To evaluate them we will quote Prandtl's BL solution of the form:

$$f(\eta) = C\exp(-\pi\eta)\cos(\pi\eta)$$
$$h(\eta) = K\exp(-\pi\eta)\sin(\pi\eta)$$

where the constants C and K are determined from the B.C's (19) and (20):

$$f(\eta) = \exp(-\pi\eta)\cos(\pi\eta) \tag{21}$$

$$h(\eta) = \frac{2\pi}{\exp(-\pi)+1}\exp(-\pi\eta)\sin(\pi\eta) \tag{22}$$

Substitution of Eqs. (21) and (22) into the expressions of the dimensionless coefficients gives:

$$\chi_0 = \pi K; \qquad \chi_1 = \pi; \qquad \chi_2 = \frac{K^2}{8\pi}\left[1-\exp\left(-2\pi\right)\right];$$
$$\chi_3 = \frac{\exp(-\pi)+1}{2\pi}; \qquad \chi_4 = \frac{K}{8\pi}\left[1-\exp\left(-2\pi\right)\right].$$

2.5.2. Dimensionless Group

Eq. (17) and (18) can now be solved since the values of all quantities except T_c and δ are known. However, our equations are still slightly complicated to be solved analytically or numerically, so we will simplify them further by introducing a dimensionless group, derived from the following order-of-magnitude analysis.

Take L_T as the vertical length scale, Eq. (11) provides the order of magnitude for u_s :

$$\dot{q} \sim \frac{\alpha W}{L_T} \Rightarrow u_s \sim \frac{\alpha W}{L_T \delta} \tag{23}$$

Balancing of the convective term and the wall conduction term in Eq. (10) gives:

$$\frac{\delta^2 u_s}{\alpha L_T} \sim 1$$

Substituting Eq. (23) into the above provides the order of magnitude for the BL thickness:

$$\delta \sim \frac{L_T^{\,2}}{W} \tag{24}$$

Now we argue that the buoyancy force is equally important as the shear force in Eq. (9):

$$g\beta\Delta T\delta \sim \frac{\nu u_s}{\delta} \tag{25}$$

where $\Delta T = T_c(z=0)$, the maximum temperature difference between the core flow and the wall. This occurs at the starting point of the BL.

Substitution of Eq. (23) and (24) into Eq. (25) gives

$$\left(\frac{L_T}{W}\right)^7 \frac{g\beta\Delta T W^3}{\alpha\nu} \approx 1$$

The term $\frac{g\beta\Delta T W^3}{\alpha\nu}$ is the product of the Grashof number, which represents the ratio of buoyancy to viscous force acting on a fluid [9], and the Prandtl number. (GrPr) forms our dimensionless group and can be substituted into Eqs. (23) ~ (25) to relate the order of magnitude of the stratified depth, the BL thickness, and the BL and core velocities to the maximum half-width:

$$\frac{L_T}{W} \approx (Gr\,\mathrm{Pr})^{-\frac{1}{7}} \tag{26}$$

$$\frac{\delta}{W} \approx (Gr\,\mathrm{Pr})^{-\frac{2}{7}} \tag{27}$$

$$u_s \approx \frac{\alpha}{W}(Gr\,\mathrm{Pr})^{-\frac{3}{7}} \tag{28}$$

$$u_{zc} \approx u_s \frac{\delta}{W} \approx \frac{\alpha}{W}(Gr\,\mathrm{Pr})^{\frac{1}{7}} \tag{29}$$

2.5.3. Scaled Variables

The final step of simplification is to scale the variables (z, T_c and δ) into dimensionless forms:

$x = \frac{z}{L_T}$, the dimensionless depth

$\theta = \frac{T_c(z)}{\Delta T}$, the dimensionless temperature. B.C: $\theta = 1$ at $x = 0$.

$\Delta = \frac{\delta W}{L_T^{\,2}}$, the dimensionless BL thickness

Substitution of these scaled variables and Eqs. (27)~(29) into Eqs. (17) and (18) gives:

$$\frac{d}{dx}\left\{\frac{1}{\Delta}\left[\frac{\theta''(x)}{\theta'(x)}\right]^2\right\} = \frac{\chi_3}{\chi_2}\Pr\theta\Delta + \frac{\chi_0}{\chi_2\sin\phi}\Pr\frac{1}{\Delta^2}\frac{\theta''(x)}{\theta'(x)} \tag{30}$$

$$\frac{d}{dx}\left[\theta'(x) - \frac{\theta\theta''(x)}{\theta'(x)}\chi_4\right] = \frac{1}{\sin\phi}\left[\chi_1\frac{\theta}{\Delta} + \cos\phi\frac{L_T}{W}\theta'(x)\right] \tag{31}$$

Eqs. (30) and (31) are the final forms of the equations that we will try to solve in the next section. Once we obtain the solutions, we will be able to plot the velocity and temperature distributions in the valley and understand the evolution of the BL.

3. Method of Solution

3.1. Initial Conditions

By observation Eqs. (30) and (31) are highly non-linear and we will need to resort to numerical analysis. This is carried out using numerical integration in MATLAB, which requires a set of initial conditions (I.C's) to initiate. We therefore need to analytically find the behaviour of the BL for very small x. In the analysis that follows, we take the limit $x \to 0$ instead of setting x = 0, because the flow becomes singular at x = 0. To examine the solution for small x, we introduce the scaled BL volume flux, *p*:

$$p = \frac{\dot{q}L_T}{\alpha W} = -\frac{\theta''(x)}{\theta'(x)} \tag{32}$$

where use has been made of Eq. (11) to eliminate $\dot{q}$. Eqs. (30) and (31) can be further simplified to:

$$\frac{1}{\Delta}p^2 + \frac{\chi_0}{\chi_2}\frac{\mathrm{Pr}}{\sin\phi}\int_0^x \frac{p}{\Delta^2}dx = \frac{\chi_3}{\chi_2}\mathrm{Pr}\int_0^x \theta\Delta dx \tag{33}$$

$$(\chi_4 - 1)\theta' p + \chi_4 \theta p' = \frac{\chi_1}{\sin\phi}\frac{\theta}{\Delta} \tag{34}$$

In effect, we have three unknowns (Δ, p, and θ) and three equations (Eqs. (32), (33), and (34)). The forms of the trial solutions are adapted from [8] to be:

$$\Delta = ax^{\frac{1}{4}} \tag{35}$$

$$p = bx^{\frac{3}{4}} \tag{36}$$

where a and b are constants. Recognising that $\frac{\theta''}{\theta'} = \frac{d}{dx}(\ln\theta')$, Eq. (32) can be integrated to give:

$$\theta' = \theta_0' \exp\left(-\int_0^x p dx\right)$$

θ_0' is the value of θ' at x = 0, which is not easily determined by analysis and will be evaluated via a set of constraints on numerical integration in Section 3.2.

Substitution of Eq. (36), integration, and application of the B.C that $(\theta)_{x=0} = 1$ gives:

$$\theta = 1 + \int_0^x \theta_0' \exp\left(-\frac{4}{7}bx^{\frac{7}{4}}\right)dx$$

For the limit of $x \to 0$ we can linearize the exponential term using power series expansion:

$$\begin{aligned}\theta &= 1 + \int_0^x \theta_0' \left(1 + O\left(x^{\frac{7}{4}}\right)\right)dx \\ &\approx 1 + \theta_0' x\end{aligned} \tag{37}$$

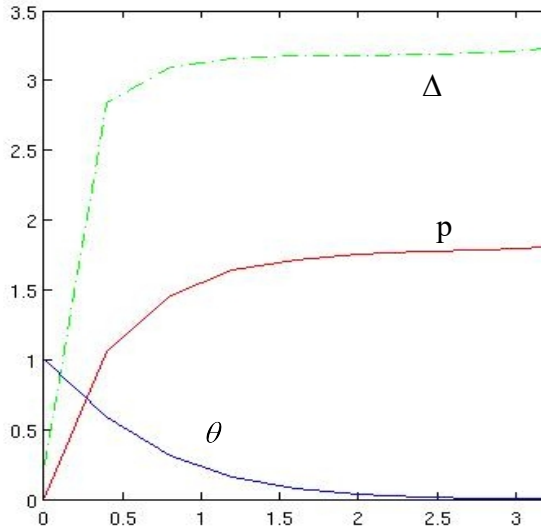
Having obtained the forms of Δ, p, and θ, we can substitute them into Eqs. (33) and (34) to calculate the constants a and b:

$$a = \left[\frac{\chi_1 \chi_2}{\chi_3 \chi_4} \frac{1}{\sin\phi} \left(\frac{5}{3} \frac{\chi_1}{\chi_4} \frac{1}{\Pr} + \frac{\chi_0}{\chi_2} \right) \right]^{\frac{1}{4}} \qquad b = \frac{4}{3\sin\phi} \frac{\chi_1}{\chi_4} \frac{1}{a}$$

Note that a and b are independent of θ_0' because the latter always appears in the equations as $\theta_0' x$, which can be neglected due to its higher order. Δ, p, and θ can now be evaluated given a small value of x to form the I.C's for numerical analysis. In our particular case, this starting value of x is set to be 1×10^{-9}.

3.2. Success of the Shooting Method for the Inviscid Case

To numerically solve Eqs. (32) ~ (34), the fourth-order Runge-Kutta method is used [10]. The three unknowns (Δ, p, and θ) are passed as elements of a single matrix to be solved by the MATLAB solver function, ode45. To gain assurance in mastering the code, I solved a pair of similar equations in Davidson's paper [8] and obtained identical solutions (Fig. 13) with the ones in the paper which were derived analytically (Fig. 14). Note that Δ on Fig. 14 is on a different scale from p and θ:



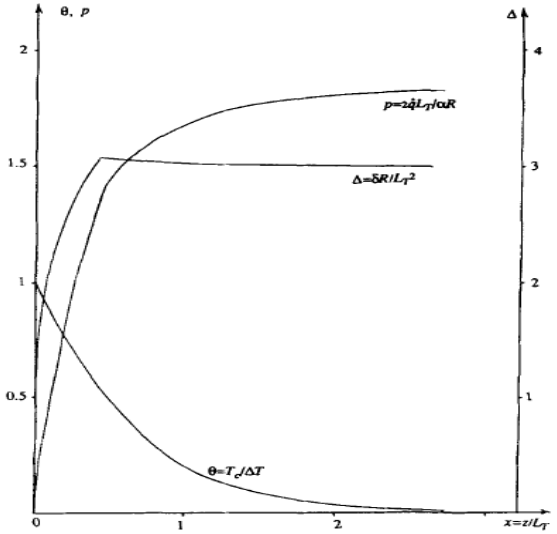


Fig. 13. Numerical Solutions **Fig. 14. Solutions by Davidson [8]**

To obtain these results we have used the shooting method [13]. Recall that the I.C of θ contains θ_0', which is unknown. The MATLAB solver function can produce solutions given a θ_0' value, but does not tell us which θ_0' to choose! We need extra constraints to find the 'correct' θ_0'.

For the above case of the Aluminum Ingot, viscosity is negligible compared to thermal diffusivity as the Prandtl number is very low. The wall jet consists predominantly of TBL and wall shear is neglected. Therefore, after some initial thermal diffusion, the wall jet reaches a steady state where temperature is homogenised and no viscosity acts to thicken the BL. In this isothermal region, the scaled BL thickness (Δ) becomes constant and so does the scaled BL mass flux (p), as indicated in Fig. 13. The 'correct' θ_0' was therefore chosen to be the one that allows p and Δ to flatten at large x.

Note that the scaled temperature difference (θ) is predicted to decay exponentially by both numerical integration and analytical solution.

The reason why the case of the Aluminum Ingot could be solved analytically is because the viscous terms were neglected taking advantage of the very low Prandtl number of Aluminum. Despite having a similar nature, our problem involves the non-linear viscous term that complicates analysis. We will next discuss the process of determining θ_0' for Eqs. (32) ~ (34).

3.3. Different Constraints for the Viscous Case

It was first attempted to choose θ_0' using the same constraints as the practise case. However, despite a wide range of θ_0' inputs, p and Δ could never flatten simultaneously. As captured by Figs. 15 and 16, p starts to decay before Δ stops growing.

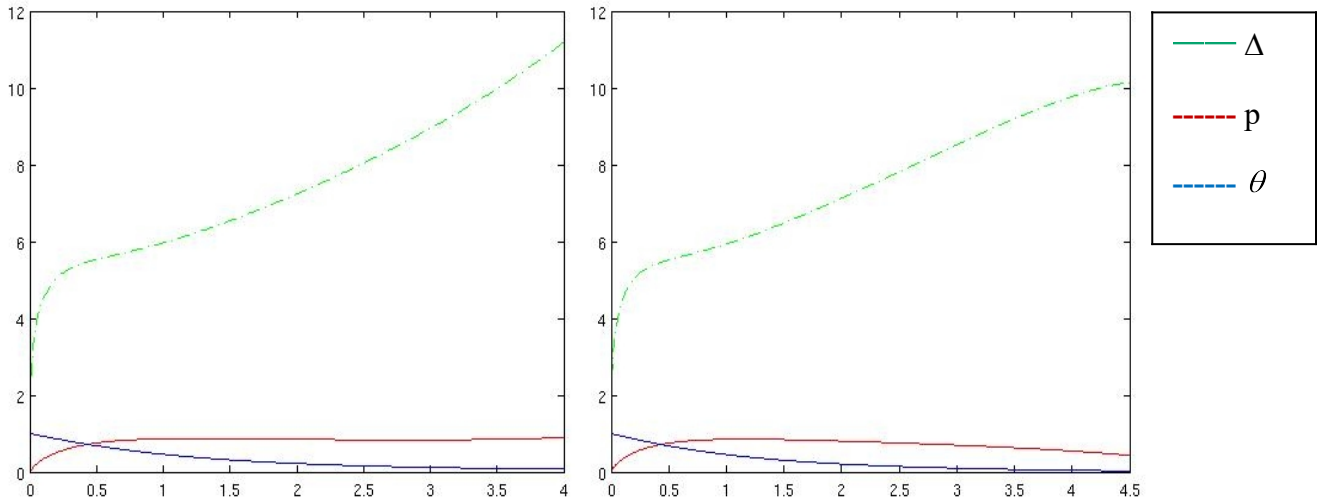


Fig. 15. (left) Growing Δ and Asymptoting p for θ_0 '= – 0.700

Fig. 16. (right) Growing Δ and Decaying p for θ_0 '= – 0.707

Therefore, although p and θ seem to evolve in a similar manner to that in the practise case, Δ behaves very differently. To investigate the reason for this we return to theoretical analysis to examine the behaviour of Δ_∞ at large x, assuming p_∞ = constant and $\theta_\infty \to 0$.

Eq. (33) rewritten in differential form gives:

$$\underset{\text{I}}{\frac{d}{dx}\left(\frac{1}{\Delta}p^2\right)} + \underset{\text{II}}{\frac{\chi_0}{\chi_2}\frac{\text{Pr}}{\sin\phi}\frac{p}{\Delta^2}} = \underset{\text{III}}{\frac{\chi_3}{\chi_2}\text{Pr}\,\theta\Delta} \qquad (38)$$

Recall the physical meaning of the three terms:

I: net rate of change of momentum

II: dissipation due to viscous stress

III: production of momentum due to buoyancy

Applying $\theta_\infty \to 0$ and p_∞ = constant in the limit of large x, Eq. (38) gives the behaviour of Δ_∞:

$$\frac{d\Delta_\infty}{dx} = \frac{\chi_0}{\chi_2}\frac{\mathrm{Pr}}{\sin\phi}\frac{1}{p_\infty} \quad \Rightarrow \quad \Delta_\infty = C_5 + \frac{C_2}{p_\infty}x \tag{39}$$

where $\mathrm{C}_2 = \frac{\chi_0}{\chi_2}\frac{\mathrm{Pr}}{\sin\phi}$ and C_5 is a constant from integration.

Therefore, for a constant p_∞, Δ_∞ does not become steady, but grows linearly with x! The reason why we were unable to obtain suitable values of θ_0' with the previous criteria has now become clear: even if p_∞ did reach a steady value, the scaled BL thickness would grow due to the viscous term which was non-exist in the model for the Aluminum Ingot.

We also need to re-examine our assumption that $\theta_\infty \to 0$. This is done by substituting Eq. (39) and p_∞ = constant into Eq. (34):

$$\frac{d}{dx}(\theta_\infty' + C_3\theta_\infty p_\infty) = C_4\frac{\theta_\infty}{\Delta_\infty}$$

$$\Rightarrow \theta_\infty = D\left(C_5 + \frac{C_2}{p_\infty}x\right)^{-\frac{C_4}{C_2(1-C_3)}}, \text{ where } C_3 = \chi_4 \text{ and } C_4 = \frac{\chi_1}{\sin\phi}.$$

Unlike the case of Aluminum Ingot where θ_∞ decays exponentially, the scaled temperature difference in our case follows a power relationship! In retrospect, the result that p_∞ = constant was necessitated by the fact that the scaled temperature difference (θ) decayed exponentially. Since $p = -\frac{\theta''(x)}{\theta'(x)}$, if $\theta(x)$ were in power form then p_∞ could never be constant, i.e. the assumption of a constant p_∞ is self-contradictory.

3.4. New Solution Forms

The above analysis is useful also because it suggests an alternative form of solution following a power relation. We can therefore define a new set of trial solutions as below:

$$\theta = D(1+\gamma x)^{-m}$$

$$p = -\frac{\theta''}{\theta'} = (m+1)\gamma(1+\gamma x)^{-1}$$

where D and m are constants to be determined.

To be compatible with p and θ, Δ needs to have the same coefficient (γ) for x:

$\Delta = E(1+\gamma x)^s$, where E and s are constants.

Substitution into Eq. (34) and matching powers and coefficients of $(1+\gamma x)$ give:

m = 7, s = 2.

We could carry on with the substitution and elimination to obtain the other constants (γ, E, and D), but the following approach is devised which is simpler and most importantly, provides a criteria for choosing θ_0'. Substitution of m and s into the trial solutions gives:

$$p = 8\gamma(1+\gamma x)^{-1} \Rightarrow \left(\frac{p}{8}\right)^{-1} = \frac{1}{\gamma} + x \tag{40}$$

$$\Delta = E(1+\gamma x)^2 \Rightarrow \Delta^{\frac{1}{2}} = E^{\frac{1}{2}} + E^{\frac{1}{2}}\gamma x \tag{41}$$

$$\theta = D(1+\gamma x)^{-7} \Rightarrow \theta^{-\frac{1}{7}} = D^{-\frac{1}{7}} + D^{-\frac{1}{7}}\gamma x \tag{42}$$

According to Eq. (40), the correct θ_0' can be determined by forcing the slope of $\left(\frac{p}{8}\right)^{-1}$ to one. The value of $\frac{1}{\gamma}$ is obtained from the interception of $\left(\frac{p}{8}\right)^{-1}$ with the y-axis. E and D are

determined from the y-interception of $\Delta^{\frac{1}{2}}$ and $\theta^{-\frac{1}{7}}$ respectively. Calculation of the slopes of $\Delta^{\frac{1}{2}}$ and $\theta^{-\frac{1}{7}}$ provides two more estimates of γ. Therefore, one estimate of γ can be obtained from each of Eqs. (40) ~ (42), which are compared to ensure that the solution is self-consistent.

3.5. Success of the Power Law

Setting the starting value of x to be 1×10^{-9} and the step size to be 3×10^{-5}, we obtain Fig. 17 which demonstrates the behaviour of the values of $\frac{1}{\gamma}$ as $\theta_0{}'$ is varied manually. Because γ is small, $\frac{1}{\gamma}$ instead of γ is compared to achieve better accuracy. Pr = 0.713 and $\phi = 90^\circ$; these two parameters will be adjusted to model various scenarios in the next section.

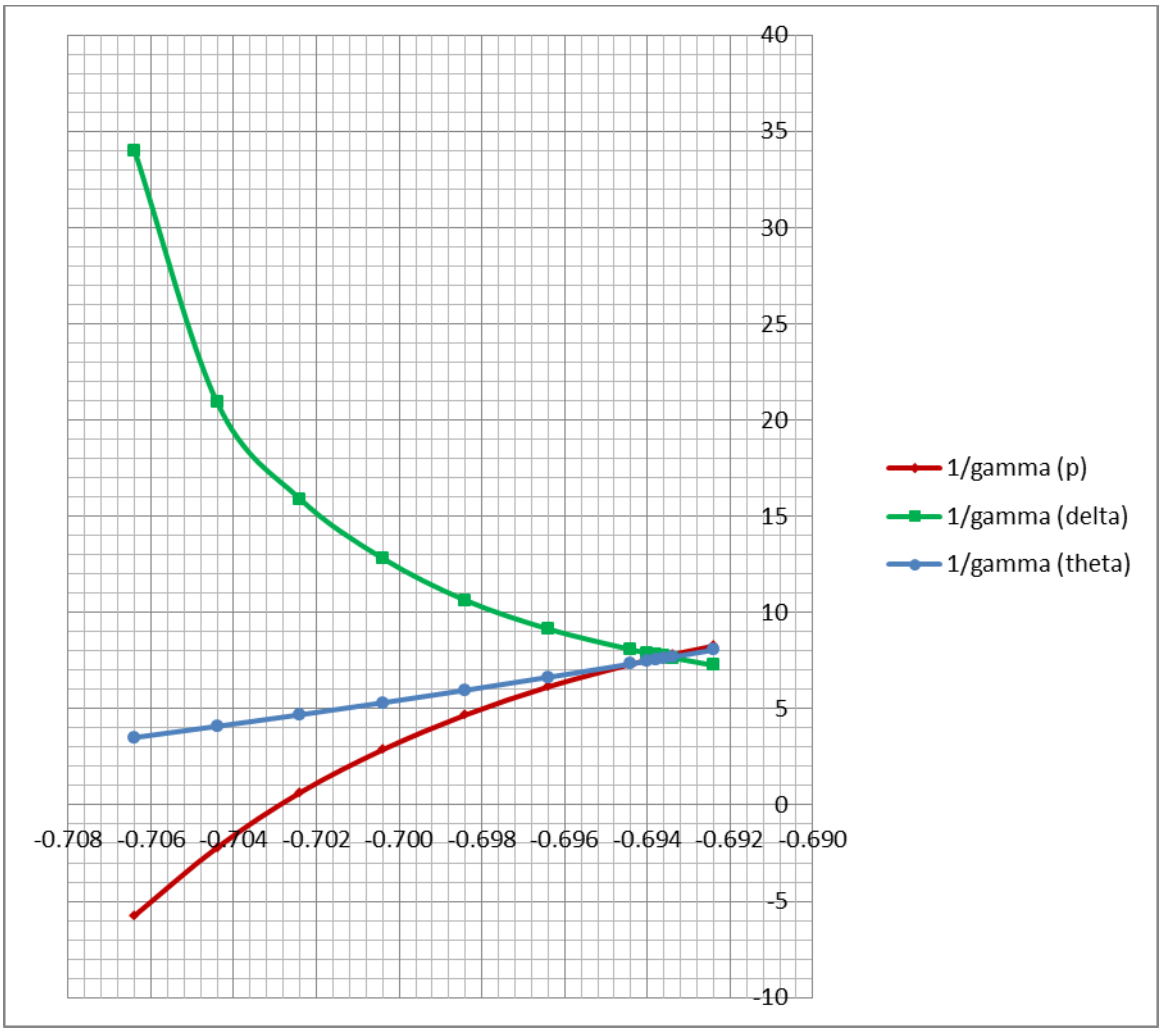


Fig. 17. Behaviour of $\frac{1}{\gamma}$ against θ_0'

The three estimates of $\frac{1}{\gamma}$ all intercept together at $\theta_0' = -0.6938$. Remarkably, at the very same value of θ_0', $\left(\frac{p}{8}\right)^{-1}$ reaches a slope of one, as shown in Fig. 18 below. Fig. 19 illustrates the application of a least squares line fit to the variables.

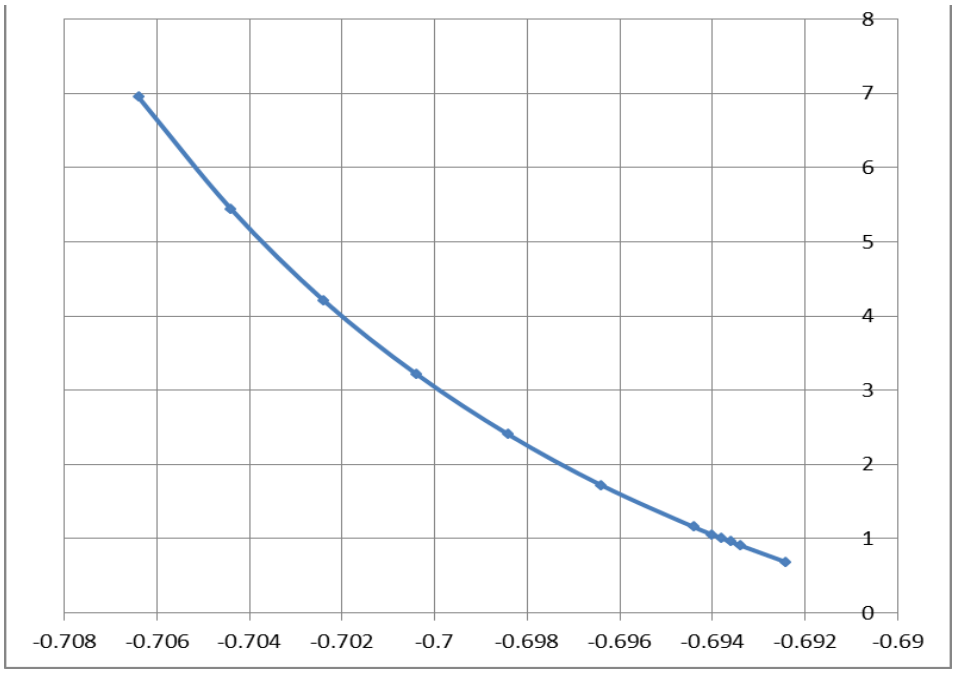


Fig. 18. Slope of $\left(\frac{p}{8}\right)^{-1}$ against θ_0'

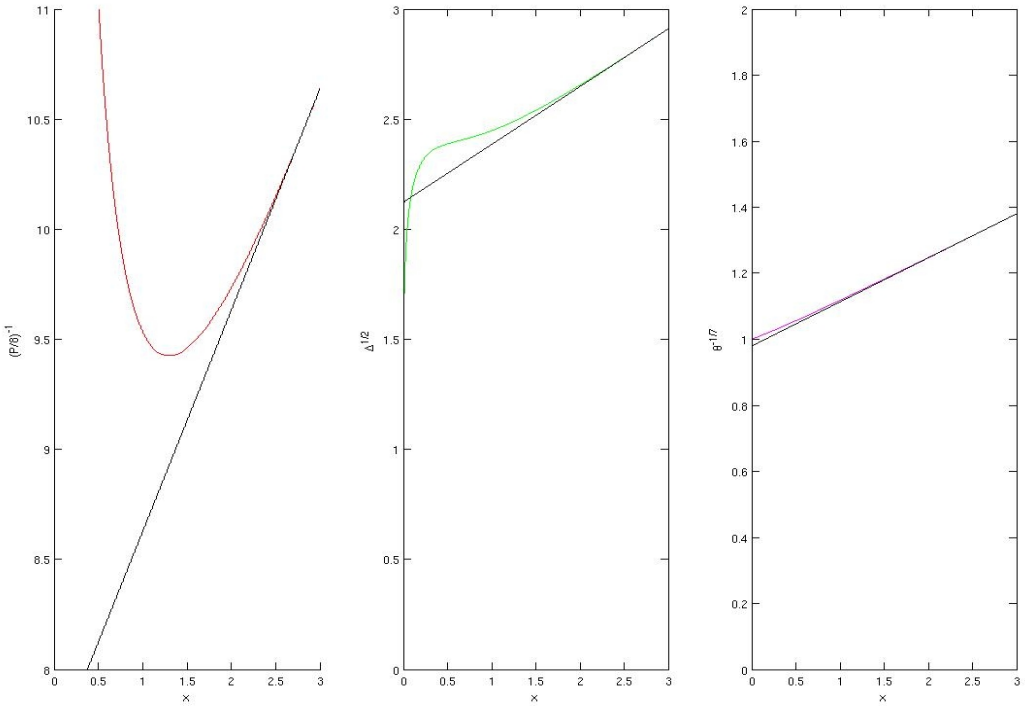


Fig. 19. Linear Line Fits of $\left(\frac{p}{8}\right)^{-1}$ (left), $\Delta^{\frac{1}{2}}$ (middle), and $\theta^{-\frac{1}{7}}$ (right)

Fig. 19 shows that the variables asymptote to their respective power forms after a depth of 1~2 L_T. This is the set of self-consistent solutions that we finally obtain. We now need to examine if these solutions are physical.

4. Physical Interpretation of Solutions

4.1. Evolution of Key Quantities

To model a realistic scenario, the valley slope chosen to be at 30° with the horizontal and the temperature for a valley at night is chosen to be near 273.15 K. At this temperature, the Prandtl number is 0.715 [11]. The shooting method subject to the power law constraints gives θ_0' = - 0.8453255. We obtain the variation of Δ, p, and θ with the scaled depth, x, in Fig. 20. Fig. 21 shows the evolution of the terms I, II, III in Eq. (38) and the BL velocity, u_s. These figures only demonstrate the relative changes of the quantities and do not reflect their physical magnitude.

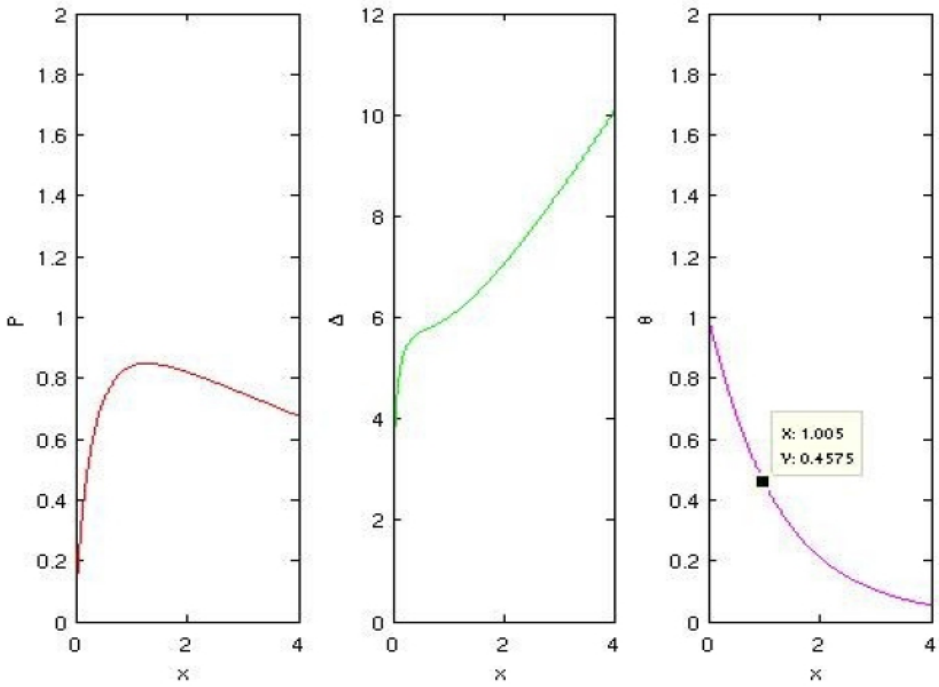


Fig. 20. Dimensionless BL mass flux (p), BL thickness (Δ), and temperature (θ) against scaled depth

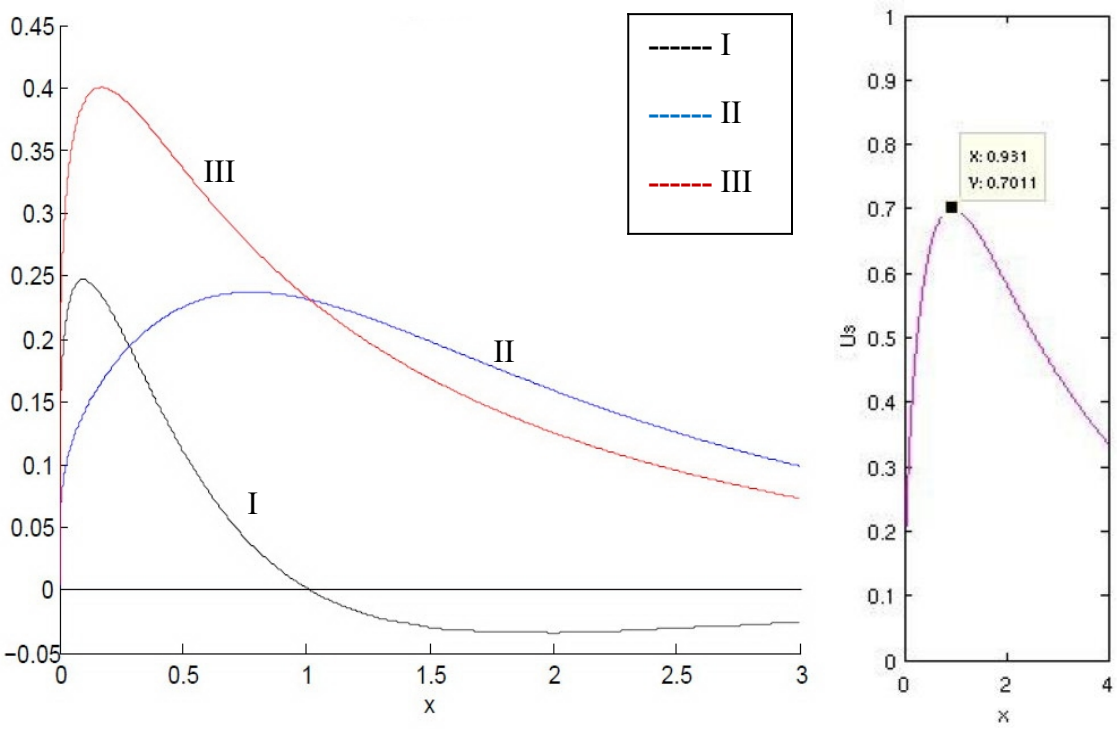


Fig. 21. Evolution of Net Change in Momentum (I), Viscous Force (II), Buoyancy Force (III), and Slope-wise BL velocity (u_s).

The above figures show that initially the wall jet has a large temperature difference (θ) and small velocity. The buoyancy force (III) is therefore large and viscous dissipation (II) is low, resulting in a net increase in momentum (I). Let us examine what happens at x=1 ($z = L_T$), roughly at the end of the stratified region. As indicated by Fig. 20, θ has fallen to less than half of its starting value as thermal diffusion homogenises temperature, causing the buoyancy force in Fig. 21 to decrease. The viscous force, on the other hand, increases within the stratified region as flow accelerates, and peaks at approximately the same time as the slope-wise BL velocity. At the end of the stratified region, buoyancy force equals viscous force and hence the net change in momentum is zero, as indicated by Fig. 21. The BL entrains core flow throughout the stratified region, causing the BL thickness in Fig. 20 to increase quickly and the BL mass flux to peak slightly after L_T.

Far away from the stratified region, temperature asymptotes to being isothermal and viscous force overtakes buoyancy. We therefore obtain an important conclusion that *within the stratified region, competition is between buoyancy and viscous forces; far away from the stratified region, competition is between viscous and inertial forces.* This leads to the continuous decay of net momentum (I) which corresponds to a fall in u_s in Fig. 21. After the temperature gradient diminishes, the TBL essentially disappears and viscosity causes the VBL to thicken and detrain back into the core. These are illustrated by the decrease of p and parabolic increase of Δ beyond x = 1 in Fig. 20.

The above analysis indicates that our solutions are compatible with the physics. It would be interesting to investigate them further by replacing all parameters with actual numbers and examining the magnitude of key quantities.

4.2. A Scenario Analysis

Our key quantities of interest are: the length scale of stratification and the Reynolds number for the core and the BL.

Let us consider a valley which is 200 meters wide (half-width = 100 m) with a surface temperature of 273.15 K at night. The ambient air is warmer than the mountain surface by 5 K at the top of the slope ($\Delta T = 5K$). At 273.15 K, the Prandtl number of air is 0.715, the laminar kinematic viscosity is $13.3\times10^{-6} m^2 s^{-1}$ and the laminar thermal diffusivity is $18.6\times10^{-6} m^2 s^{-1}$ [11]. These give a laminar Grashof number of 1.015×10^{15}.

However, in natural convection, transition to turbulent flow takes place in the range $10^8 < Gr < 10^9$; above this range, the BL is turbulent [9]. Moreover, recall Eq. (26) which links L_T to the dimensionless group (GrPr), a laminar Grashof number will give $\frac{L_T}{W} = (Gr\,\mathrm{Pr})^{-\frac{1}{7}} = 7.534\times10^{-3}$, which means that for a 200-meter wide valley, the stratified depth is only one meter - way too small!

Therefore the flow has to be turbulent. Due to the time constraint for this project, however, we can only reasonably model turbulence by keeping our model and replacing the kinematic viscosity with an eddy viscosity (ν_{turb}). From Eq. (26), the eddy viscosity is determined by $\frac{L_T}{W}$, which is estimated following the analysis below.

If $\frac{L_T}{W} = 0.2$, the definition of Δ gives the BL thickness $\delta = \Delta \frac{L_T^{\,2}}{W} = 4\Delta$. As Δ grows to 10 at z = 80 m (Fig. 20), the BL thickness becomes 40 m, which is about half of W and twice the magnitude of L_T. This is apparently too thick and L_T needs to be smaller to scale δ down. For $\frac{L_T}{W} = 0.1$, $\delta = \Delta$ and becomes 10 m at z = 40 m, which is acceptable. We therefore arrive at the estimate that $\frac{L_T}{W} \approx 0.1$.

Substitution of this ratio into Eq. (26) gives the turbulent Grashof number and the eddy viscosity. Assuming that the Prandtl number remains constant at 0.715, the turbulent thermal diffusivity can be obtained:

$$(Gr\,\mathrm{Pr})^{-\frac{1}{7}} = 0.1 \Rightarrow Gr = 1.399\times10^{7},\ \nu_{turb} = 0.1133,\ \alpha = 0.1585$$

We can now estimate the BL velocity along the slope from the BL mass flux:

$$\dot{q} = -\alpha w \frac{T_c''(z)}{T_c'(z)} = -\alpha w \frac{p}{L_T} \Rightarrow$$

$$u_s = -\frac{\dot{q}}{\delta} h(\eta) = \frac{w}{L_T}\frac{\alpha p}{\Delta} h(\eta) \approx 1.2 ms^{-1}$$

This enables us to evaluate the Reynolds number for the BL: $\mathrm{Re}_{BL} = \frac{u_s \delta}{\nu} \approx 6\times10^{5}$, which is in the turbulent regime. Note that the slope angle (ϕ) does not enter into any expressions above and hence this estimate applies to valleys of any slope.

The core velocity and Reynolds number are calculated using the core equation:

$$u_{zc} = -\alpha \frac{T_c''(z)}{T_c'(z)} = \alpha \frac{p}{L_T} \approx 0.01 ms^{-1}$$

$$\mathrm{Re}_{core} = \frac{u_{zc} W}{\nu} \approx 9.5 \times 10^4$$

In contrast to most planar BL's in the engineering course, the core velocity for our case is much smaller than the BL velocity. This is a direct result of continuity $\left(u_{zc} \approx u_s \frac{\delta}{W} \right)$ and the fact that the width of the valley is much greater than the thickness of the BL.

4.3. Re-examination of Assumptions

With the above values we are able to re-examine the assumptions we made in Section 2.2 when proving core thermal stratification. These are: $\delta \le L_T$ and $L_T \le W$. The latter is automatically satisfied from the ratio we chose in Section 4.2.

For $z \le L_T$, $\frac{\delta}{L_T}$ ranges from 0 to 0.5 and hence our assumption is valid within the stratified region. Beyond this depth, the core becomes isothermal. Appendix III explains mathematically why $\frac{\delta}{L_T}$ could reach a large value within the isothermal region.

The above estimates enable us to input real numbers to MATLAB and plot our solutions, as discussed in Section 4.4.

4.4. Velocity and Temperature Distributions

Converting the slope coordinate into Cartesian, Fig. 22 is obtained showing core and wall jet velocity vectors for $\phi = 30^{\circ}$. The core flow travels upwards, then turns to enter the BL where kinetic energy is dissipated and heat is lost to the mountain surface. While inside the BL, the flow travels downslope under density effects before exiting the BL to join the core again. It is observed that the maximum velocity in the valley is concentrated in the wall jet, which matches our theoretical analysis that $u_{zc} < u_s$. The BL entrains core flow as it accelerates and then detrains as it decelerates. The core velocity is mostly vertical and relatively small, except in regions close to the wall and to the top. Near the top, the horizontal velocity component dominates so that streamlines curve and enter the BL, as predicted in Solution B to Eq. (3).

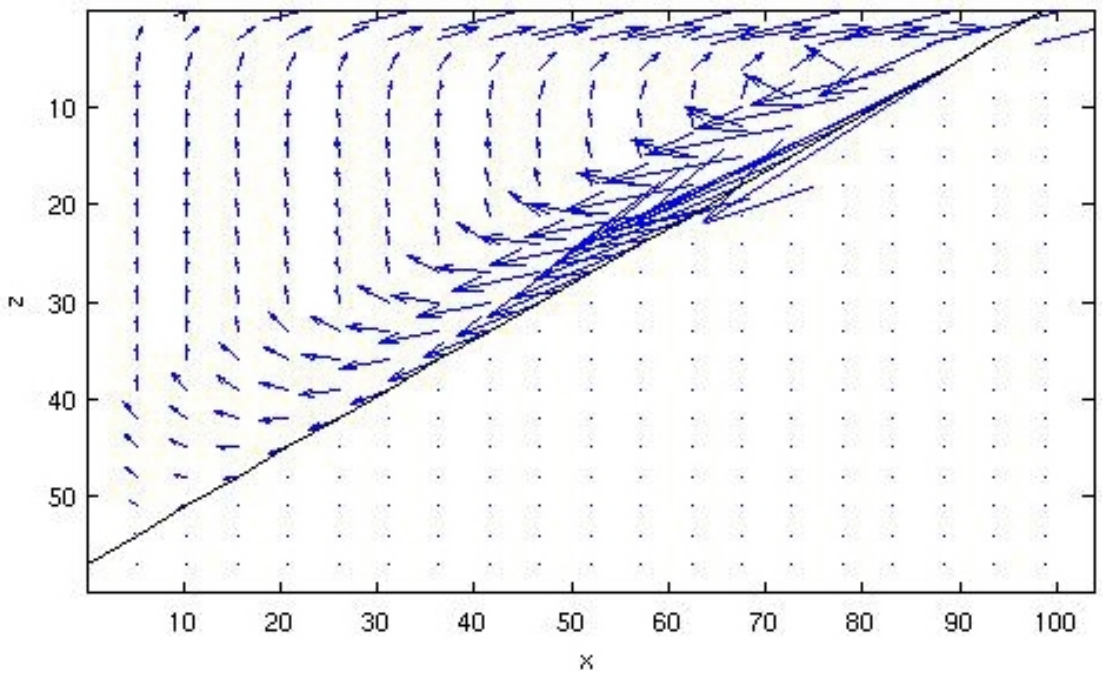


Fig. 22. Velocity Vector Plot for $\phi = 30^{\circ}$

The acceleration and deceleration of the wall jet are better illustrated with Fig. 23. For clarity, core vectors are erased and only velocities inside the BL are shown. Fig. 24 is an enlargement of Region A in Fig. 23.

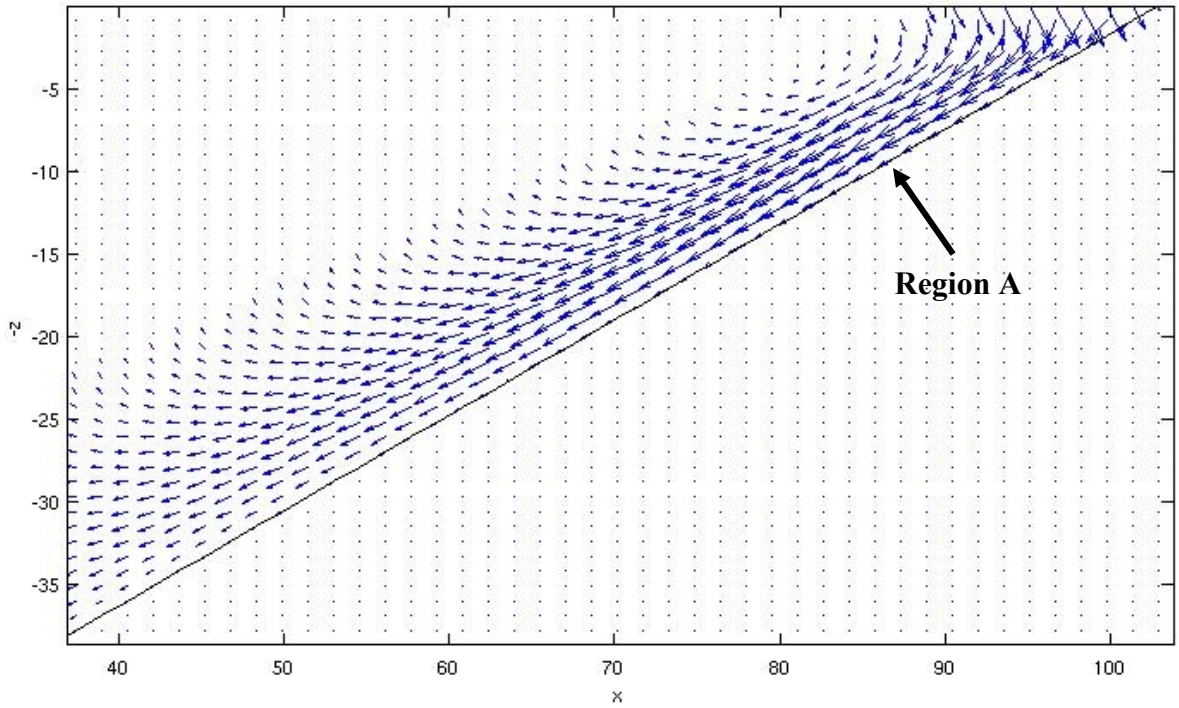


Fig. 23. Vector Plot showing BL velocities only for $\phi = 30^{\circ}$

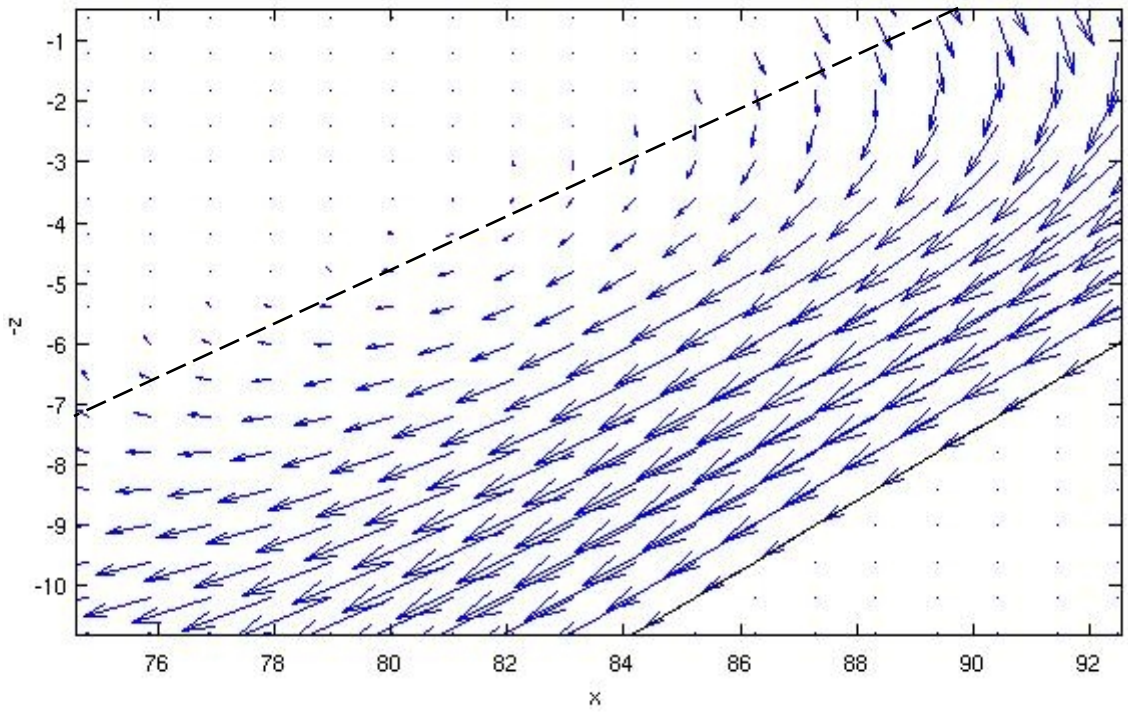


Fig. 24. Enlargement of Region A in Fig. 23

The dashed line in Fig. 24 indicates the BL boundary. In this figure, the initial entrainment and later detrainment of the core flow are visualised by the vectors crossing this boundary. Recall that the typical depth of the stratified region is 10 m. Velocity vectors are observed to enter the BL for z < 5 m and leave towards the end of the stratified region. The magnitude of the velocity vectors diminishes towards both the BL boundary and the wall due to the form of the velocity profile function (Eq. (22)): $h(\eta) = \frac{2\pi}{\exp(-\pi)+1}\exp(-\pi\eta)\sin(\pi\eta)$.

The temperature distribution of the core and the BL is plotted as shown in Fig. 25. Each coloured line is an isotherm at the temperature specified on the spectrum to the right.

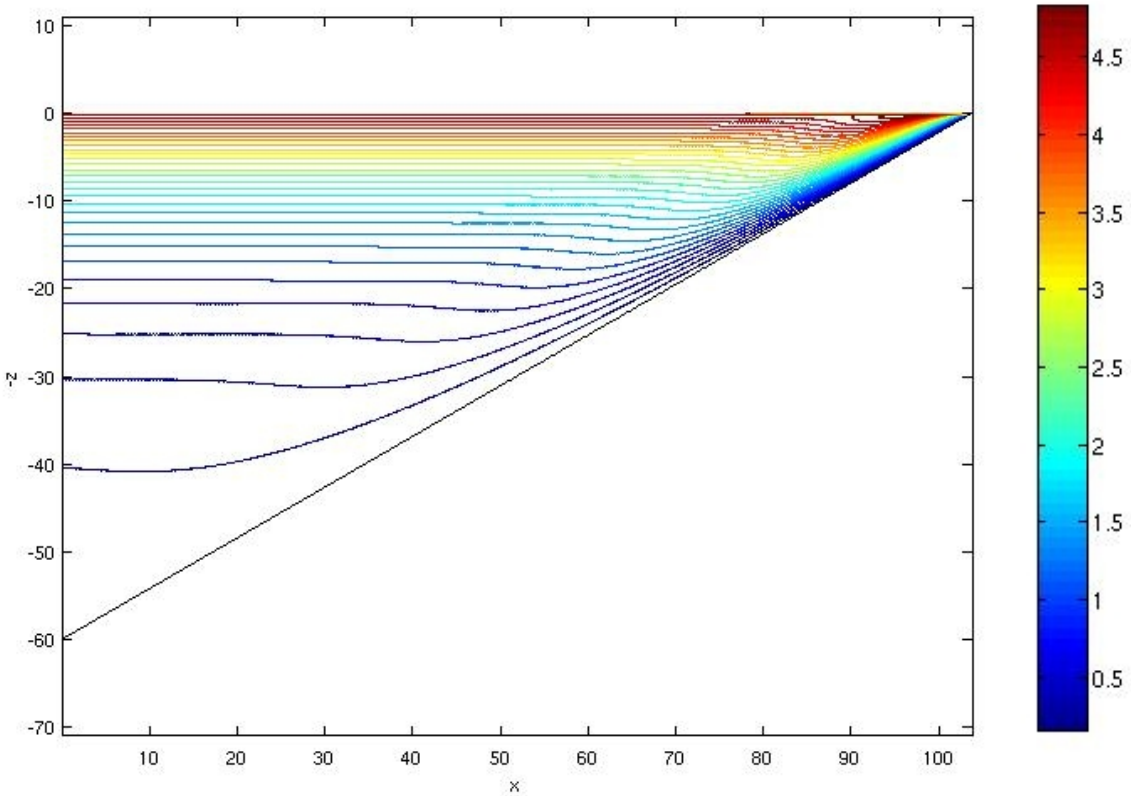


Fig. 25. Isotherms for $\phi = 30^{\circ}$

Temperature (relative to T_{wall}) ranges from 5 (red) to 0 (blue) and isotherms that are close together indicate a steep temperature gradient. The core temperature exhibits strong stratification for $z \le L_T$ as indicated by many clustered isotherms near the top. Deeper down the valley, the temperature gradient becomes more and more mild and beyond z = 40 m, the

core becomes approximately isothermal. The isotherms inside the BL curve upwards, mathematically due to the form of the temperature profile function (Eq. 21) and physically because temperature drops close to the mountain surface.

4.5. Effect of a Larger Slope Angle

We will investigate the effect of changing the slope angle on flow pattern and temperature distributions. At a steeper angle of $\phi = 60^{\circ}$, θ_0' = - 0.722541535 from manual testing. We obtain the vector plot (Fig. 26) and the temperature plot (Fig. 27).

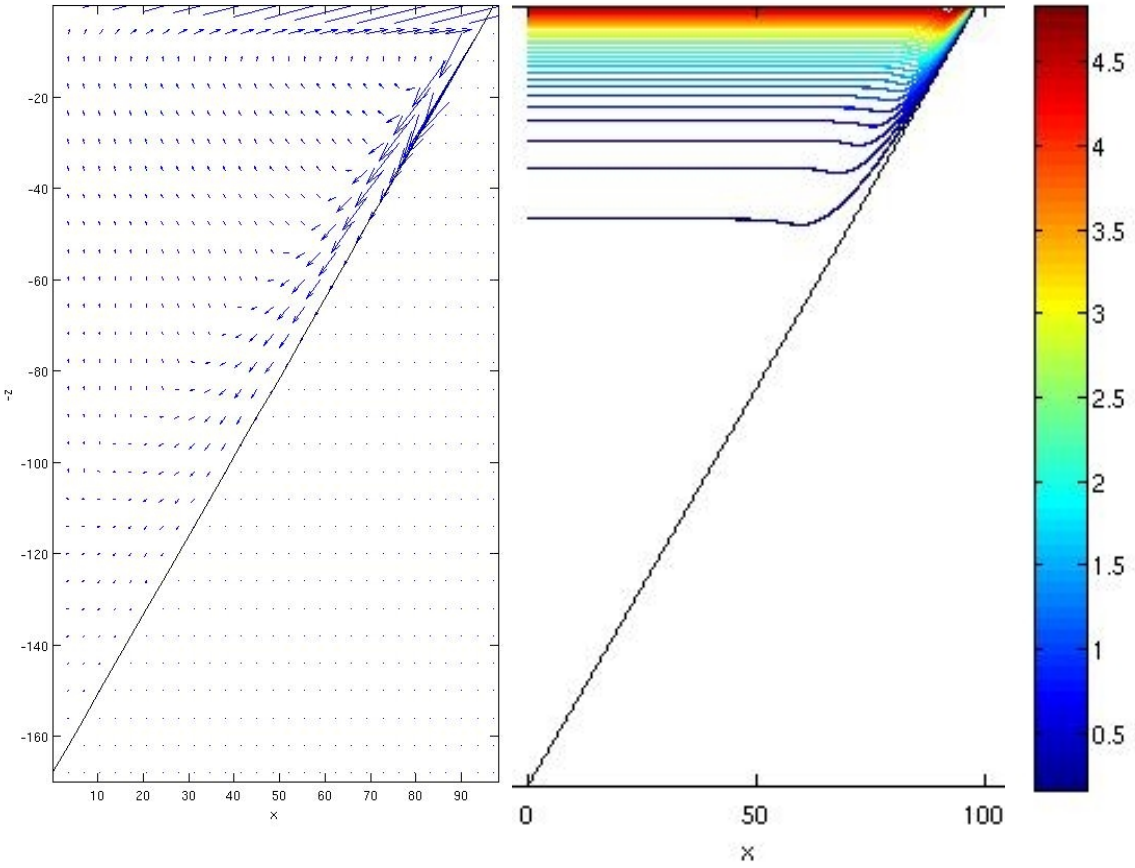


Fig. 26. Vector Plot for $\phi = 60^{\circ}$

Fig. 27. Isotherms for $\phi = 60^{\circ}$

Cross comparison of Fig. 26 and Fig. 27 leads to the conclusion that *the largest velocities are inside the BL across the depth where thermal stratification is significant.* This feature is also common in Fig. 22 and is explained treating temperature gradient as the source of kinetic energy.

It is observed that the isothermal region for $\phi = 60^{\circ}$ is larger than that for $\phi = 30^{\circ}$, because for a fixed width, a steeper slope angle means a deeper valley where thermal diffusion takes place over a longer distance. However, for both slope angles temperature becomes homogeneous at around the same absolute depth of 40 m.

To examine the behaviour of the wall jet, we obtain Fig. 28 which shows an enlargement of the top region (z < 40 m) in Fig. 27. Fig. 29 is an enlargement of Region B in Fig. 28.

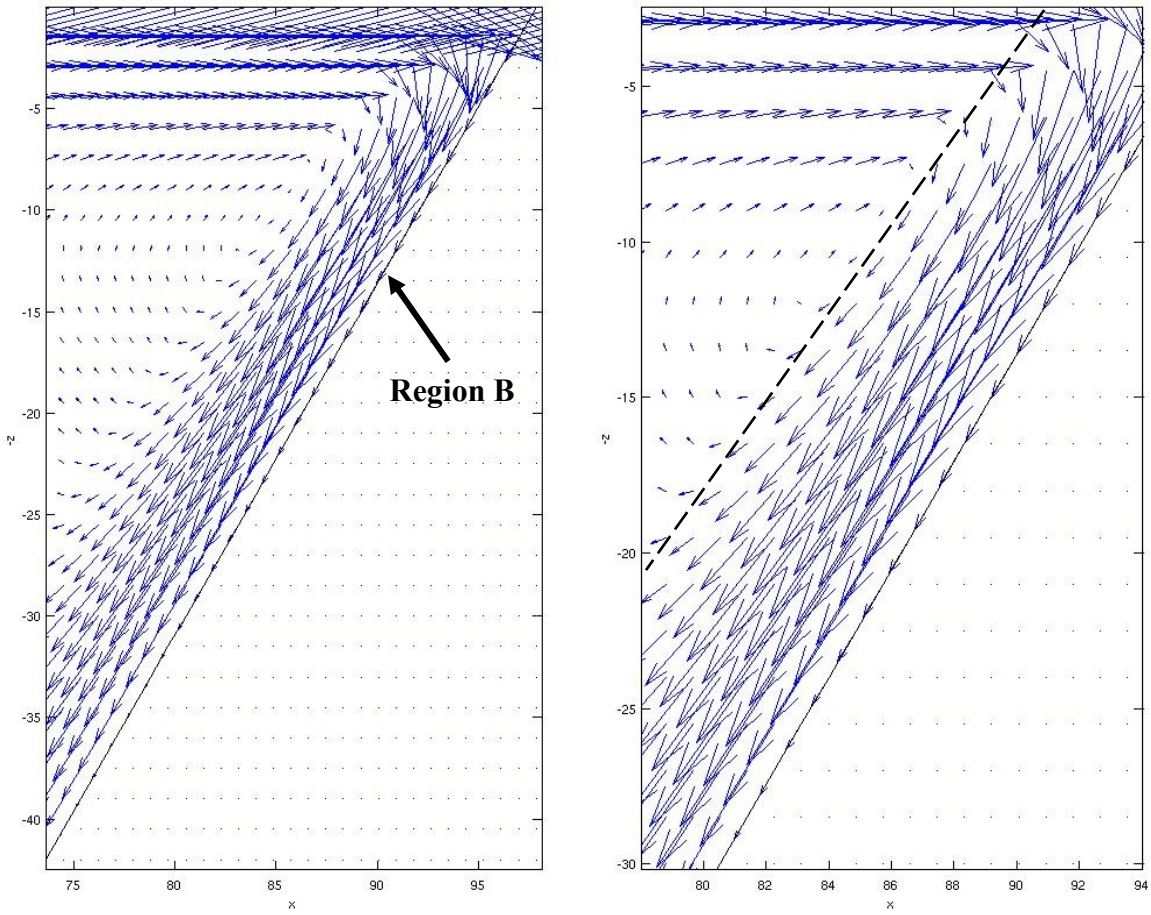


Fig. 28. (Left) Enlargement of the Top Region (z < 40 m) in Fig. 27

Fig. 29. (Right) Enlargement of Region B in Fig. 28

Due to a greater depth, the BL thickness could in theory grow to 50 m at the bottom of the valley! This is a direct result of the power law which predicts a parabolic growth of BL thickness with depth (Fig. 20). However, in reality this is limited to the width of the valley, which decreases linearly with depth. At a depth of around 120 m, the flow consists entirely of VBL with no core flow.

Fig. 29 exhibits similar patterns of entrainment and detrainment with those at $\phi = 30^{\circ}$. However, the depth at which detrainment initiates is approximately 8 m, compared to 5 m at the smaller angle. This is because at a larger angle, the length of the slope across the same stratified region is shorter, as indicated by Fig. 30 where $s_{60} < s_{30}$. This means that for a fixed stratified depth, viscous shear dissipates energy over a shorter distance and therefore the point at which deceleration starts is delayed.

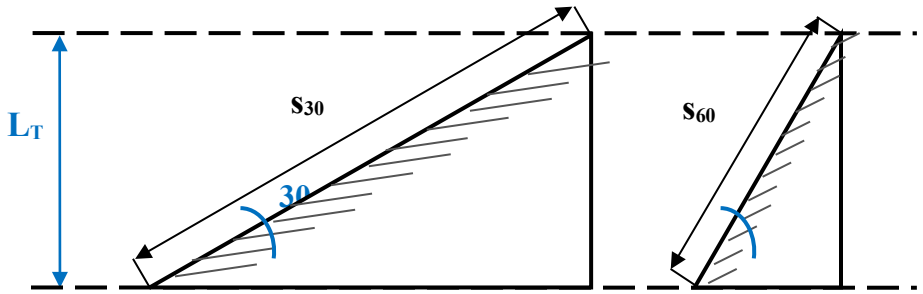


Fig. 30. Shorter Slope at a Larger Angle across the Stratified Depth

Apart from the above differences, the flow remains largely unchanged. The evolution of the rate of change of momentum (I), viscous force (II), and buoyancy force (III) in Eq. (38) is similar at both angles, as indicated by Fig. 31 and Fig. 32. Note that the horizontal axis is the dimensionless depth, $\frac{z}{L_T}$.

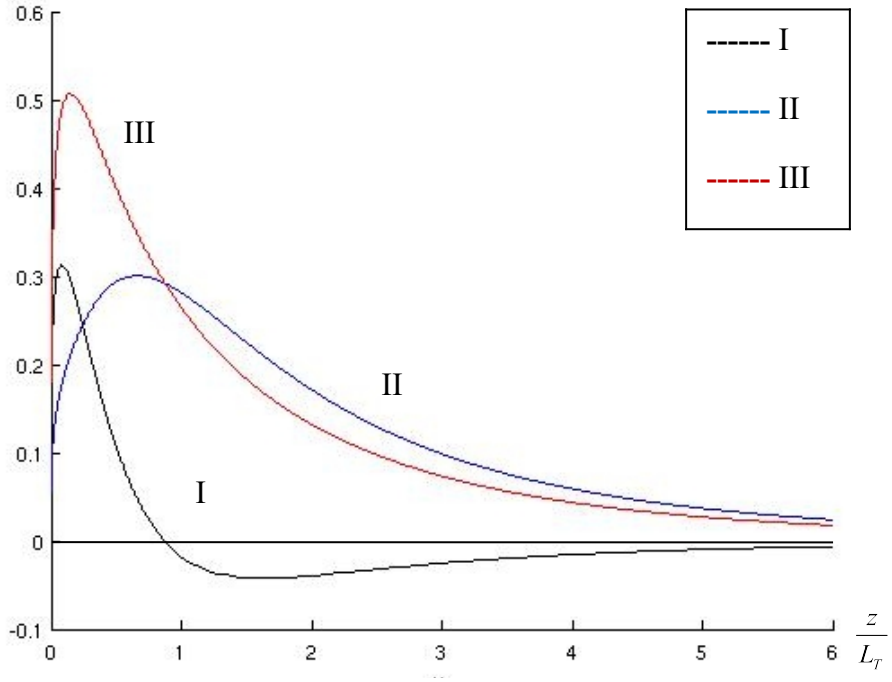


Fig. 31. Evolution of Net Change in Momentum (I), Viscous Force (II), and Buoyancy Force (III) for $\phi = 30^{\circ}$

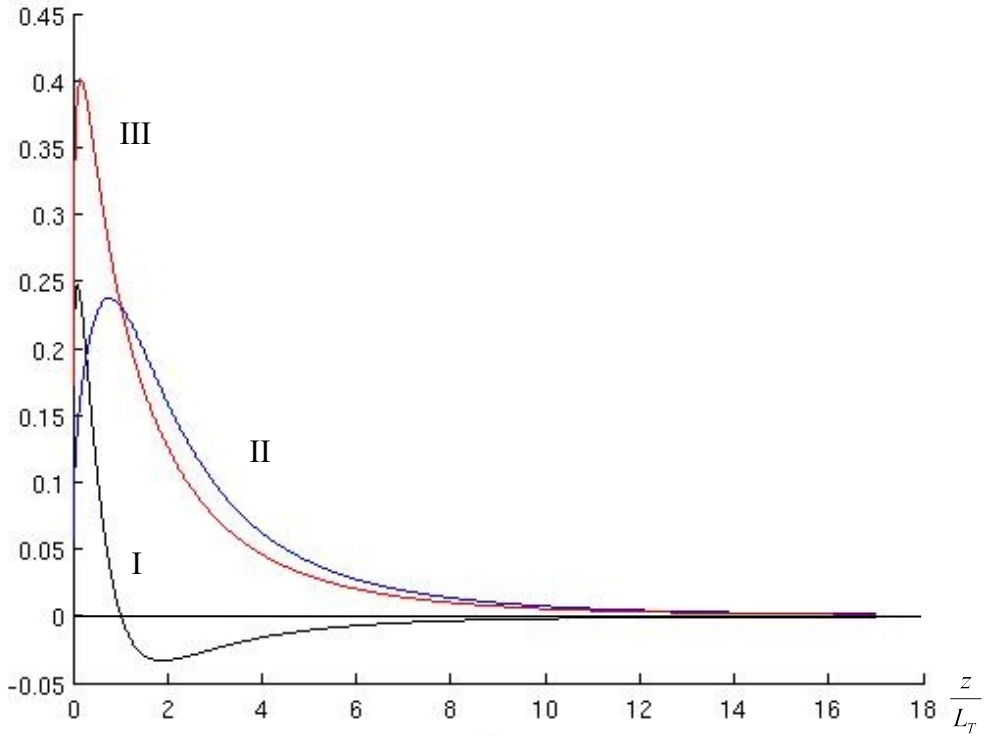


Fig. 32. Evolution of Net Change in Momentum (I), Viscous Force (II), Buoyancy Force (III) for $\phi = 60^{\circ}$

Both Fig. 31 and Fig. 32 show the initial domination of buoyancy force and the decay of all terms at large depth. Fig. 31 looks more compressed in the x-direction due to the greater range of depth plotted. Recall the expression of the three terms in Eq. (38):

$$\underbrace{\frac{d}{dx}\left(\frac{1}{\Delta}p^2\right)}_{\text{I}}+\underbrace{\frac{\chi_0}{\chi_2}\frac{\Pr}{\sin\phi}\frac{p}{\Delta^2}}_{\text{II}}=\underbrace{\frac{\chi_3}{\chi_2}\Pr\theta\Delta}_{\text{III}}$$

The viscous term (II) is the only one that contains the slope angle. The factor of $\frac{1}{\sin\phi}$ arises from resolving the vertical depth to the distance along the slope. This confirms our previous argument and leads to the conclusion that *increasing the slope angle has the effect of diminishing the magnitude of the viscous term relative to the buoyancy term, therefore increasing the depth at which detrainment takes place.*

5. Future Work

5.1. Improvements to Existing Model

Despite being able to produce overall sound results, our analysis adopts a turbulence model that is oversimplified. The following three alternatives are suggested which provide better turbulence modelling.

In his analysis of an infinite heated plate, Prandtl gave a model of the shear stress for the turbulent case: $\tau = \zeta \rho u_s^{\ 2}$, where ζ is a resistance coefficient that varies with the degree of turbulence. The wall shear stress is then equated with the order of magnitude of the thermal thrust on the TBL: $\zeta \rho u_s^{\ 2} \approx \rho g \beta \sin\phi C \delta$, where C is a constant. The thermal diffusivity is connected with turbulence through a turbulence exchange coefficient equal to $\zeta \delta u_s$, obtained from an order-of-magnitude analysis. This approach is entirely analytical.

Alternatively, the following method can be used which involves both analytical and numerical analysis. For the turbulent case, the time-averaged Navier-Stokes Equation and heat transport equation include correlation terms:

$$\left(\overline{\mathbf{u}} \cdot \nabla\right)\overline{\mathbf{u}} = -\nabla\left(\frac{P}{\rho}\right) - g\overline{\beta}\overline{T}\mathbf{k} + \nu\nabla^2\overline{\mathbf{u}} - \left(\mathbf{u}' \cdot \nabla\right)\mathbf{u}' - g\beta' T'\mathbf{k}$$

$$\left(\overline{\mathbf{u}} \cdot \nabla\right)\overline{T} = \alpha\nabla^2 T - \left(\mathbf{u}' \cdot \nabla\right)T'$$

where the apostrophe symbolizes fluctuations around mean quantities.

Introduction of these correlation terms into Eqs. (3) and (4) would largely complicate the proof of core stratification. For simplicity, we can assume that the flow is turbulent in the BL but remains laminar in the core due to a smaller velocity. Steady thermal stratification in the core also helps to suppress turbulence. A slight complication associated this approach is that we need to use the turbulent viscosity and thermal diffusivity for the BL, but use the laminar ones when analysing the core flow.

A more ambitious approach would be to approximate the correlation terms using the $k-\varepsilon$ model throughout the valley. In this case, we will not divide the flow into a core and a BL, but will instead solve all equations numerically. If numerical treatment capacity allows, it would also be interesting to examine the evolution of the Reynolds shear stress, the mean flow kinetic energy, and the turbulent kinetic energy.

5.2 Further Investigations

If given extra time, I would also like to investigate the case of the open ended valley (recall Fig. 4) and compare it with the results for the blocked valley. For this case, the core is quiescent and we only need to solve our turbulent momentum and heat transport equations for the BL.

It is proposed to conduct on-site experiments to measure the actual temperature and wind velocity. This would help to improve our various assumptions. For example, we assumed the temperature of the mountain surface to be constant, while Prandtl assumed it to be at a fixed temperature below the local core flow. This is determined by the thermal diffusivity of the mountain material, which is a ratio of thermal conductivity to specific heat capacity [12]. Materials with a large thermal diffusivity tend to conduct heat quickly relative to its heat capacity and hence changes its temperature fast to reach thermal equilibrium, vice versa. Site measurements would provide a suitable means to decide how to model the mountain surface temperature.

6. Conclusions

We will summarize the general findings of this project:

- Under steady state, the flow inside a two-dimensional valley consists of a thermally stratified core and a boundary layer.
- All core streamlines pass through the boundary layer where kinetic energy produced by buoyancy is dissipated.
- Inside the boundary layer within the stratified region, competition is between buoyancy and viscous forces; beyond the stratified region, competition is between viscous and inertia forces.
- Entrainment of the core flow into the boundary layer takes place within the stratified region, followed by detrainment as the flow becomes approximately isothermal.
- The largest velocities in the valley are confined to the boundary layer across the depth where thermal stratification is significant. In most part of the core which is far away from the wall and from the top, core velocity is relatively small.
- Increasing the slope angle has the effect of increasing the proportion of the core that is isothermal and increasing the thickness of the boundary layer at the bottom of the valley.
- For a fixed depth of the stratified region, a greater slope angle also decreases the distance across which viscous force dissipates energy, causing flow detrainment to take place at a larger depth.

7. Acknowledgements

With thanks to Prof. Davidson who shed light whenever there is darkness and Dr. Agarwal and Dr. Liang who offered helpful feedback on my work.

8. References

[1] A. Bajić, B. Peroš, V. Vučetić, Z. Žibrat, Wind load - a meteorological basis for Croatian standards, GRAĐEVINAR 53 (2001), page 501

[2] NASA Earth Observatory. URL: http://earthobservatory.nasa.gov

[3] Gesellschaft, *VDI Heat Altas* (2nd ed.). Springer Reference 2010, pp. 12

[4] L. Prndtl, *Essentials of Fluid Dynamics, With Applications to Hydraulics, Aeronautics, Meteorology and other subjects*, 1952, pp. 422-224.

[5] Encyclopedia Britannica, Inc, "Mountain Breeze: Valley and Mountain Wind Dynamics", 2008. URL: http://www.britannica.com/EBchecked/media/111214/When-the-valley-floor-warms-during-the-day-warm-air

[6] MetEd, "Katabatic Winds". URL: https://www.meted.ucar.edu

[7] Integrated Publishing, "Drainage Winds". URL: http://www.tpub.com/weather2/3.htm

[8] P.A. Davidson, *Natural Convection in an Aluminium Ingot: A Mathematical Model*, Metallurgical and Materials Transactions, Volume 25 B, April 1994 - 295, pp. 3, 7.

[9] M. Lappa, *Thermal Convection, Patterns, Evolution, and Stability*, 2010, p. 66

[10] Süli, Endre; Mayers, David (2003), *An Introduction to Numerical Analysis*, Cambridge University Press, ISBN 0-521-00794-1, p. 328

[11] Engineering Toolbo,x "Air Properties". URL: http://www.engineeringtoolbox.com/air-properties-d_156.html

[12] Lide, David R. (2009). *Handbook of Chemistry and Physics*. Boca Raton, Florida: CRC Press. p. 2-65. ISBN 978-1-4200-9084-0.

[13] *Numerical Recipes: The Art of Scientific Computing* (3rd ed.). New York: Cambridge University Press. ISBN 978-0-521-88068-8.

9. Appendices

Part I.

Proving $g\beta \oint_C T dz = 0$ if $T = T(z)$:

$$g\beta \oint_C T dz = g\beta \oint_C T\mathbf{e}_z \cdot d\mathbf{r} = g\beta \iint_S \nabla \times (T\mathbf{e}_z) \cdot d\mathbf{A}$$

where Stokes' Theorem has been applied to obtain the last expression. We now invoke the vector identity:

$$\nabla \times (T\mathbf{e}_z) = T\nabla \times \mathbf{e}_z + (\nabla T) \times \mathbf{e}_z$$

$$g\beta \iint_S \nabla \times (T\mathbf{e}_z) \cdot d\mathbf{A} = g\beta \left[\iint_S T\nabla \times \mathbf{e}_z \cdot d\mathbf{A} + \iint_S \nabla T) \times \mathbf{e}_z \cdot d\mathbf{A} \right]$$

$\nabla \times \mathbf{e}_z \cdot d\mathbf{A} = 0$ since $\nabla \times \mathbf{e}_z$ is normal to $d\mathbf{A}$:

$$g\beta \oint_C T d\mathbf{z} = g\beta \iint_S \nabla T) \times \mathbf{e}_z \cdot d\mathbf{A}$$

Since $T = T(z)$, $\nabla T = T'(z)\mathbf{e}_z$ and therefore $(\nabla T) \times \mathbf{e}_z = 0$.

Part II.

Proving that if all streamlines go through the BL, then $T = T(z)$:

Taking the curl of Eq. (1) gives the transport equation for vorticity:

$$\frac{D\omega_y}{Dt} = \frac{\partial}{\partial x}(g\beta T) + \nu\nabla^2\omega_y$$

Inside the core viscous diffusion is negligible. The advection of vorticity and buoyancy production are hence the major forces and have the same order of magnitude:

$$\mathbf{u}\cdot\nabla(\omega_y) \sim g\beta\frac{\partial T_c}{\partial x} \tag{7}$$

where $\omega_y = \frac{\partial u_{xc}}{\partial z} - \frac{\partial u_{zc}}{\partial x}$ (8)

Applying continuity to the core gives:

$$\frac{\partial u_{xc}}{\partial x} + \frac{\partial u_{zc}}{\partial z} = 0 \Rightarrow u_{zc} \sim u_{xc}\left(\frac{L_T}{W}\right)$$

This allows us to compare the relative sizes of the two terms in Eq. (8):

$\frac{\partial u_{zc}}{\partial x} \sim u_{xc}\left(\frac{L_T}{W^2}\right) \sim \frac{\partial u_{xc}}{\partial z}\left(\frac{L_T}{W}\right)^2 \ll \frac{\partial u_{xc}}{\partial z}$, assuming that $\frac{L_T}{W} \ll 1$.

We return to justify this assumption in Section 4.3 and merely state that it is true here.

We now have $\omega_y \approx \frac{\partial u_{xc}}{\partial z} \sim \frac{u_{xc}}{L_T}$, which can be substituted into the advection term of Eq. (7) to give:

$$\mathbf{u}\cdot\nabla(\omega_y) \approx u_{xc}\frac{\partial}{\partial x}\left(\frac{\partial u_{xc}}{\partial z}\right) + u_{zc}\frac{\partial}{\partial z}\left(\frac{\partial u_{xc}}{\partial z}\right)$$
$$\sim \frac{{u_{xc}}^2}{WL_T} + \frac{u_{zc}u_{xc}}{{L_T}^2} \sim {u_{zc}}^2\left(\frac{W}{{L_T}^3}\right)$$

Elimination of u_{zc} using Eq. (5) and Eq. (6) gives another expression for $\mathbf{u}\cdot\nabla(\omega_y)$:

$${u_{zc}}^2\left(\frac{W}{{L_T}^3}\right) \sim {u_{zb}}^2\left(\frac{\delta^2}{W{L_T}^3}\right) \sim g\beta\frac{T_c\delta^2}{W{L_T}^2}$$

Equating this to the RHS of Eq. (7) gives $g\beta\frac{\partial T_c}{\partial x} \sim g\beta\frac{T_c\delta^2}{W{L_T}^2} \Rightarrow \frac{\partial T_c}{\partial x} \sim \frac{T_c}{W}\left(\frac{\delta}{L_T}\right)^2$

Provided that $L_T \ll \delta$, T_c is a function of z only because $\frac{\partial T_c}{\partial x} \to 0$. This can also be seen from $T_c = T(z)\left\{1 + O\left[\left(\frac{\delta}{L_T}\right)^2\right]\right\} \approx T(z)$.

Part III.

The reason why $\frac{\delta}{L_T}$ could reach a value greater than one could be explained using the dimensionless group relations. Recall Eqs. (26) and (27):

$\frac{L_T}{W} \propto (Gr\,\mathrm{Pr})^{-\frac{1}{7}}$, $\frac{\delta}{W} \propto (Gr\,\mathrm{Pr})^{-\frac{2}{7}}$

$\Rightarrow \frac{\delta}{L_T} = C(Gr\,\mathrm{Pr})^{-\frac{1}{7}}$, where C is a constant of order 1.

For laminar flow $(Gr\,\mathrm{Pr}) \sim 10^{15}$ and $C(Gr\,\mathrm{Pr})^{-\frac{1}{7}} \ll 1$. However, after the turbulent adjustment, $(Gr\,\mathrm{Pr}) \sim 10^{7}$, $(Gr\,\mathrm{Pr})^{-\frac{1}{7}} \sim 10^{-1}$ and the effect of the constant C becomes significant. For C > 10, $\delta > L_T$.